

Causal Fibres: Structural Adaptation for Partial Transformer Distillation

Mathematical Foundations, Representation Contests, Predictive-Coding Residuals, Continual Learning, and Software Architecture

A comparative framework for cap-amenable, partially error-predictive-coding networks

Technical review of the `causal-fibres` framework

July 2026

Abstract

This report presents the mathematical, algorithmic, experimental, and software architecture embodied in `causal-fibres`, a PyTorch-first research toolkit for comparing structured adaptation systems and for building partially predictive-coding or error-predictive-coding residual networks on top of existing neural models. The initial use case is partial Transformer distillation: preserve a competent backpropagation-trained Transformer as a frozen or lightly co-adapted substrate, then learn a smaller residual system that may support sparse or modular representations, transient predictive-coding settlement, selective continual-learning plasticity, and a registered neural-symbolic cap.

A central scientific principle is that orthogonal causal fibres should not be assumed to be the natural representation of a trained Transformer. They are one hypothesis in a matched-budget representation contest that also includes dense bottlenecks, LoRA-style substrate adaptation, sparse autoencoders, grouped and possibly overlapping overcomplete dictionaries, and context-conditioned fibre bundles. A frozen backbone is treated first as an observability control, not as a permanent doctrine. Exact autograd credit with responsibility-conditioned plasticity is the conservative default; a learned causal highway is an optional later hypothesis that must demonstrate held-out transfer, locality, or computational value. Predictor-only, free, constrained, and teacher-clamped settlement are evaluated separately so that oracle information is never confused with deployable performance.

The report develops five mathematical layers. First, teacher-gap closure formalizes whether a proposed frozen read/write interface is observable. Second, representation classes are compared through reconstruction, steering, sparsity, intervention selectivity, and budget metrics. Third, state-PC and damped Gauss–Newton error predictive coding (ePC) settlement are formulated in small residual coordinates with explicit gauge control and deployment-honest amortization. Fourth, causal-fibre emergence is analyzed through common invariant subspaces, overcomplete dictionary coordinates, context signatures, perturbation gaps, intervention fingerprints, and context-dependent transports. Fifth, continual-learning behavior is analyzed through off-support gradient leakage, mixed Hessian coupling, local commutators, and direct finite curriculum-order tests.

The report also reviews the complete package architecture: representation models, observability probes, architecture adapters, residual sidecars, solvers, exact-gradient plasticity gates, optional error highways, operator audits, interventions, lifecycle and certification machinery, symbolic template stores, staged experiment recipes, checkpoints, and reproducibility tools. Sandbox examples are used throughout, including the unsuccessful full-network activity-energy-monotonicity distillation experiment, frozen-sidecar proofs of mechanism, causal-credit misallocation tests, robust joint block diagonalization, representation contests under superposition, frozen-interface observability,

low-rank substrate co-adaptation, honest settlement-mode comparisons, and finite curriculum-order evaluation. The final chapters specify a falsifiable GPU program for partially distilling Transformers into cap-amenable, partially ePC-structured networks without presupposing that any one representation paradigm will win.

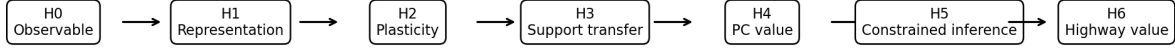
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Later mechanisms advance only after earlier empirical gates pass.

Figure 1: The evidence-gated hypothesis ladder embodied in the framework. Later mechanisms should advance only after earlier empirical gates pass.

Executive summary

`causal-fibres` is best understood as a *comparative structural-adaptation laboratory*. It asks a sequence of increasingly strong questions rather than assuming at the outset that a Transformer contains clean orthogonal causal modules.

Hypothesis	Required empirical question
H_0	Does the proposed read/write interface expose the desired correction at all?
H_1	Which representation class best captures the correction at matched capacity and compute?
H_2	Can selective persistent plasticity reduce unrelated drift at matched current-task performance?
H_3	Do learned supports and intervention fingerprints generalize to unseen contexts?
H_4	Does state-PC or ePC settlement add value beyond matched feed-forward or recurrent controls?
H_5	Does constrained settlement exploit genuinely new test-time information?
H_6	Does a learned highway improve locality, transfer, or cost relative to exact gated credit?

The safe default is deliberately conservative:

Frozen base + ordinary dense residual control + exact autograd credit + no settlement + mandatory observability and SAE baselines.

Every stronger mechanism must earn its place.

The package compares five main representation families:

$$\begin{aligned}
 \text{Dense residual:} & \quad z = E(h), \quad \hat{r} = Wz, \\
 \text{Sparse autoencoder:} & \quad h \approx Dz, \quad m = \dim z > d, \quad z \text{ sparse}, \\
 \text{Grouped sparse dictionary:} & \quad z \text{ sparse with possibly overlapping groups } S_q, \\
 \text{Orthogonal fibres:} & \quad \mathcal{H} = U_1 \oplus \cdots \oplus U_Q, \\
 \text{Contextual bundle:} & \quad z_{\text{local}} = T_c z_{\text{canonical}}.
 \end{aligned} \tag{1}$$

Orthogonal fibres are attractive when a family of context-conditioned operators admits an approximately common block structure. Sparse overcomplete dictionaries are attractive when the network stores more features than hidden dimensions through superposition. Contextual bundles are attractive when causal or functional content is stable but its local coordinates rotate with context, layer, modality, or task. The honest output of an experiment may therefore be any of the following:

Possible conclusion	Interpretation
Orthogonal fibres supported	A common direct-sum structure is empirically useful and stable.
Overcomplete grouped dictionary wins	Superposed sparse features explain the correction better.
Contextual transports required	Stable content exists only through context-dependent local charts.
Functional but not causal modules	Interventions are useful but do not identify a unique real-world mechanism.
Frozen interface unobservable	The selected reads do not expose the teacher correction.
Repeated or underidentified	Multiple modules are observationally or interventionally interchangeable.
No useful modular structure found	Dense adaptation is currently the better model.

Predictive coding enters only after representation and support controls are credible. A small residual state may be initialized by an amortized predictor and refined by state-gradient PC or a damped Gauss–Newton ePC update:

$$z^{k+1} = z^k - \eta M_k^{-1} \nabla_z E(z^k). \quad (2)$$

This can improve optimization, recurrent computation, robustness, or the use of additional constraints. It does not create new information if it sees only the same frozen features as the predictor. Accordingly, the package reports four modes separately:

$$\text{predictor,} \quad \text{free settlement,} \quad \text{constrained settlement,} \quad \text{teacher-clamped diagnostic.} \quad (3)$$

For continual learning, exact gradient credit is initially retained. Responsibility estimates gate *persistent plasticity*:

$$\nabla_{\phi_q} L_c \leftarrow r_{cq}^p \nabla_{\phi_q} L_c. \quad (4)$$

A learned causal highway is tested later and only if it generalizes credit or support to novel contexts, reduces a real computation or locality bottleneck, or serves a learning regime where exact reverse-mode gradients are unavailable.

The main target architecture remains a partial-distillation sidecar:

$$F_S(x) = F_0(x) + R_\phi(h^0(x), c), \quad (5)$$

with optional low-rank co-adaptation of selected base modules and optional transient residual states. This creates an experimentally accessible locus for continual adaptation and neural-symbolic constraints without requiring immediate full-network conversion.

Evidence status. The report combines several evidence levels. The finite-dimensional operator and perturbation results are mathematical statements under explicit assumptions. The causal-continual-learning analysis is primarily local and small-step unless finite curriculum tests are run. The sandbox examples are controlled CPU demonstrations rather than natural-language benchmarks. The software package has automated tests and runnable examples, but remains an alpha research toolkit. The proposed Transformer GPU program is a falsifiable design, not yet an empirical result.

1 Scope, terminology, and notation

1.1 Predictive-coding terminology

Throughout this report, *ePC* means *error predictive coding*: predictive-coding inference and learning formulated in prediction-error coordinates. In linear predictive-coding analyses, the error-to-state Jacobian can whiten the state Hessian, preserve the same equilibrium, and accelerate settlement. The practical residual implementation uses small error-coordinate states together with Fisher or Gauss–Newton metrics, damping, and trust-region control.

The report compares ordinary state-PC, ePC, damped Gauss–Newton settlement, natural-gradient state inference, and feed-forward or recurrent controls explicitly. The unsuccessful six-layer study discussed below is described as an *activity-energy-monotonicity distillation experiment*; it is not used to define ePC.

1.2 Basic notation

Symbol	Meaning
F_0	competent frozen backbone or base network
F_T	teacher network or teacher behavior
h_ℓ^0	frozen hidden state at architecture site or layer ℓ
a^0	frozen base logits or final output coordinates
z_q	canonical latent state of causal fibre q
B_q	fibre write map into a local or terminal representation
$A_{\ell q}$	read map from frozen site ℓ to a fibre-specific observation
$T_{\ell q}$	transport from canonical fibre coordinates to site-local coordinates
g_{cq}	activation or plasticity gate for context c and fibre q
H_c	context-conditioned Fisher, Gauss–Newton, Hessian, or update operator
P_q	orthogonal projector onto fibre subspace U_q
e_j	error component indexed by candidate causal mechanism j
r_{qj}	causal responsibility assigned from error component j to fibre q
N_j	novelty or residualization operator for mechanism j
M_q	fibre metric or precision used in state settlement
ϵ_g	gradient mass outside a context’s intended support
ϵ_h	mixed Hessian coupling across intended support boundaries

2 Evidence-gated scientific design

2.1 What the earlier evidence did and did not show

The earlier positive sandbox results established a conditional mechanism:

$$\text{known fibres} + \text{known supports} + \text{good coordinates} \implies \text{clean context-local adaptation.} \quad (6)$$

They did not establish that useful causal fibres emerge automatically from realistic Transformer states. This distinction changes the purpose of the software. The package is not a machine for assuming fibres and then confirming them. It is an instrument for deciding whether a useful structure is orthogonal, overcomplete, context dependent, merely functional, underidentified, or absent.

The most important negative diagnostic remains that energy descent and task improvement do not certify causal correctness. In a rotated-coordinate sandbox, a sparse update recovered nearly all one-step energy improvement while placing more than half its update outside the canonical causal support. A second important diagnostic is that teacher-clamped settlement may look excellent while predictor-only deployment remains mediocre. A third is that repeated isomorphic modules may admit numerically perfect block fits while being intrinsically non-identifiable.

2.2 Why superposition changes the representation question

A fixed orthogonal direct sum

$$\mathcal{H} = U_1 \oplus \cdots \oplus U_Q \quad (7)$$

is only one possible representation model. A superposed Transformer may instead admit an overcomplete sparse dictionary

$$h \approx Dz, \quad D \in \mathbb{R}^{d \times m}, \quad m > d, \quad (8)$$

with sparse code z . Functional or causal groups may overlap:

$$\mathcal{F}_q = \{j : j \in S_q\}, \quad S_q \cap S_r \neq \emptyset. \quad (9)$$

A third possibility is a contextual bundle:

$$z_{c,\ell} = T_{c,\ell} z_{\text{canonical}}, \quad (10)$$

where stable content is represented in different local charts. The framework therefore makes SAE plus steering a mandatory rival and treats joint block diagonalization as a sufficient structural hypothesis rather than the universal linear algebra of Transformers.

2.3 Freezing as a falsifier, not a doctrine

A frozen base answers the question: does the existing representation expose enough information for a small residual to do useful work? Let h_ℓ^0 be a proposed read site and let R be an ordinary trained residual. Teacher-gap closure is

$$\text{Closure} = 1 - \frac{\text{KL}(p_T \| p_{0+R})}{\text{KL}(p_T \| p_0)}. \quad (11)$$

If this control fails, a PC or fibre failure is uninterpretable. The interface, read sites, write path, or substrate policy must change.

Freezing may also be the bottleneck. The framework therefore permits controlled low-rank co-adaptation:

$$W = W_0 + \frac{\alpha}{r}BA, \quad (12)$$

subject to an anchor trust region such as

$$L_{\text{anchor}} = \lambda [D(f_{\text{current}}, f_{\text{anchor}}) - \rho]_+^2. \quad (13)$$

The scientific comparison is frozen residual versus low-rank co-adaptation versus ordinary LoRA and full fine-tuning, not frozen residual versus nothing.

2.4 The amortization boundary

If free settlement sees only the same deterministic frozen features as the predictor, then it has no information-theoretic advantage over a sufficiently expressive feed-forward map. PC may still supply a useful training target, recurrent compute, optimization bias, or robustness, but deployment value must be compared at matched parameters and compute.

Test-time settlement has a genuine informational role only when it receives additional constraints or observations:

$$z^* = z^*(h^0, m_{\text{retrieved}}, r_{\text{symbolic}}, z_{\text{previous}}, o_{\text{new}}). \quad (14)$$

This is why constrained settlement is separated from free settlement and why a symbolic or memory cap must eventually demonstrate functional inferential work rather than merely retrieval accuracy.

2.5 Why exact gradients remain the default

On a small sidecar, exact autograd credit is inexpensive and reliable. Selective continual learning can be tested by applying responsibility to plasticity while retaining exact credit. A highway is valuable only if it solves a separate problem: unavailable gradients, hardware locality, asynchronous learning, biological locality, lower cost, or transfer of a learned credit function to unseen contexts.

The decisive highway experiment is therefore held out by context: train routing and transport on one family of contexts, then compare exact gradient, exact gradient plus learned gate, dense learned feedback, and learned causal highway on unseen compositions or domains.

2.6 Identifying variation as the main scale-up risk

Synthetic grammar supplies exact interventions and minimal pairs. Natural language rarely does. The evidence ladder should proceed from exact synthetic interventions to semi-synthetic natural language, human-validated minimal pairs, controlled prompt or environment variation, and finally observational corpora. In the last regime the package should use weaker claims such as “context-stable functional module” unless stronger causal assumptions are justified.

A useful empirical separation score is

$$Z_{qr} = \frac{\|\hat{\lambda}_q - \hat{\lambda}_r\|}{\text{SE}(\hat{\lambda}_q - \hat{\lambda}_r)}. \quad (15)$$

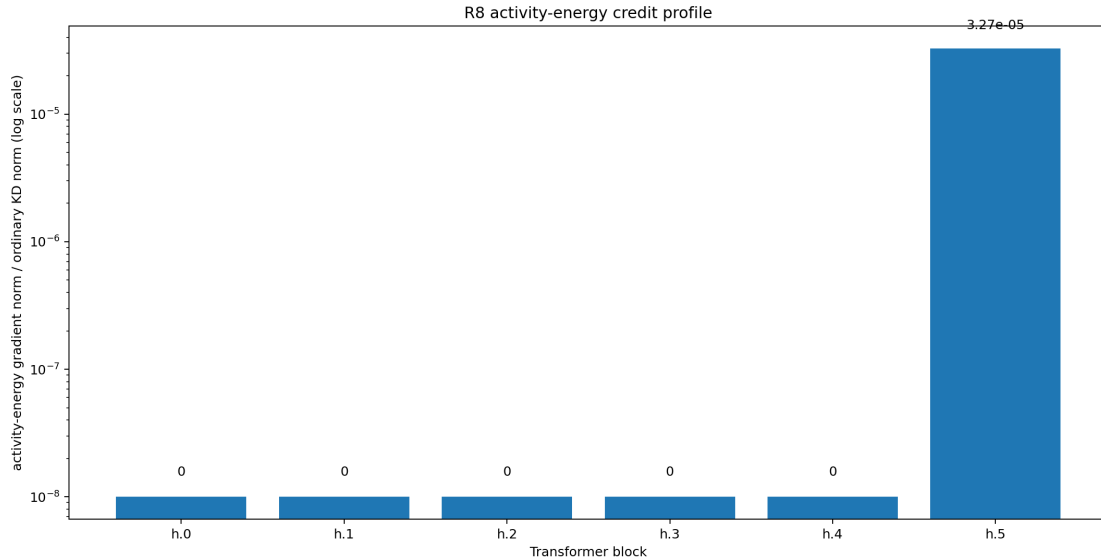


Figure 2: Credit profile reported in the original six-layer distillation experiment. Zero entries are plotted at a visualization floor. The useful point is not a modest attenuation but an almost complete concentration of the activity-energy signal in the final block.

When Z_{qr} is small, regularization cannot manufacture identification.

2.7 Local commutators versus finite curricula

For small gradient steps, Lie brackets and Hessian commutators diagnose leading-order order dependence. Real training contains long nonlinear sequences. The framework therefore measures complete curriculum maps directly:

$$D_{\text{order}}(\pi, \sigma) = D_f(U_{\pi_T} \cdots U_{\pi_1} \theta_0, U_{\sigma_T} \cdots U_{\sigma_1} \theta_0). \quad (16)$$

The practical scientific question is whether local leakage and commutator metrics predict finite order sensitivity, not whether a two-step expansion should be treated as a global guarantee.

3 Why a frozen residual sidecar was proposed

3.1 The original full-network result

The motivating experiment distilled a GPT-2-small teacher into a six-layer GPT-2 student on a synthetic subject–verb–object grammar. At 200 matched updates, the activity-energy-monotonicity arms did not beat the ordinary cross-entropy baseline; at matched wall-clock time, ordinary KD ran many more updates and achieved substantially better validation loss. More diagnostically, the activity-energy-objective gradient was numerically zero in blocks 0–4 and nonzero only in block 5, where its norm was about 3.3×10^{-5} of the ordinary KD gradient norm.

The same checkpoints showed a strong representational bottleneck. Final entropy effective rank was about 13.9 for the cross-entropy baseline, about 5.0 for ordinary short-budget KD, and about 3.8–3.9 for the activity-energy arms. The collapse was not neuron death; rather, variance concentrated into a very small subspace. Deeper iterative inference slightly intensified the contraction.

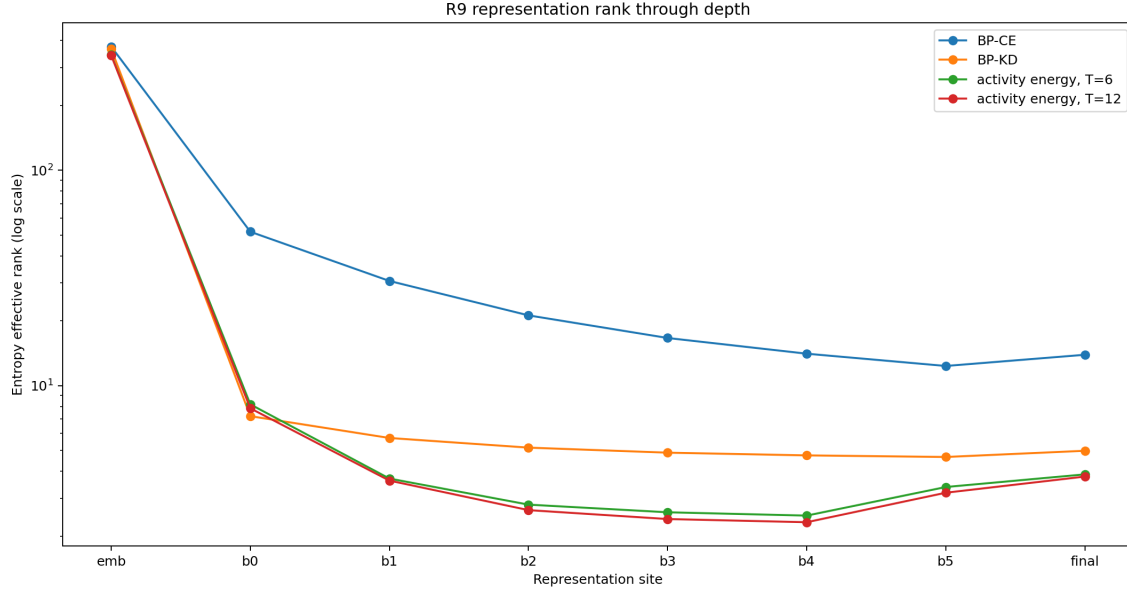


Figure 3: Entropy effective rank through depth in the R9 representation battery. The log scale makes the rapid collapse after the embedding layer visible.

Factor probes and intervention selectivity deteriorated in parallel. Subject-number probe accuracy was near chance at block 3 for the activity-energy arms, and causal selectivity fell much faster than in the cross-entropy model.

3.2 Mechanistic interpretation

The result is consistent with two distinct failures.

1. **Credit starvation.** The structural energy constraint did not deliver a task-relevant learning signal to early or middle blocks.
2. **Cheap contraction.** The objective could be satisfied by reducing representational degrees of freedom rather than by discovering cleaner causal organization.

A causal gate multiplying a near-zero early-layer signal would remain near zero. Strong sparsity or clarity penalties applied before competence could make contraction cleaner rather than make causal factors emerge. The safer response is to isolate the unconventional learning dynamics in a small residual whose failure cannot erase the base representation.

3.3 The lower-risk route

The proposed route is:

1. retain a competent frozen Transformer;
2. read selected frozen hidden states;
3. train a small state-PC/ePC/causal-coding residual sidecar;

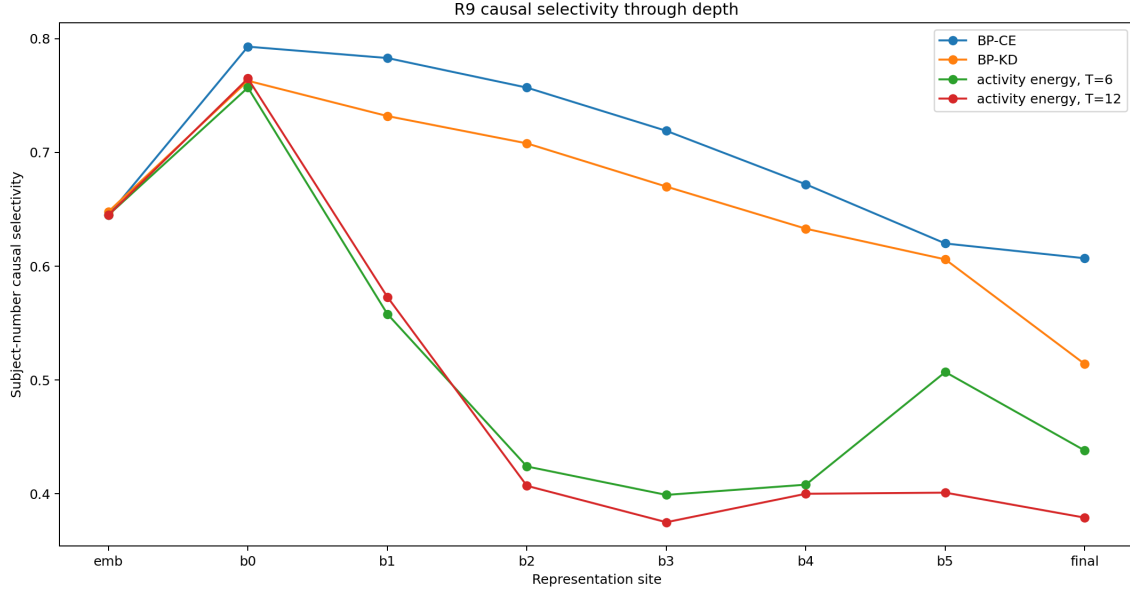


Figure 4: Subject-number causal selectivity through depth in the R9 battery. The input representations begin similarly, but the activity-energy conditions lose selectivity rapidly.

4. make the causal fibres and symbolic slots live in that sidecar;
5. use the sidecar as the clean interface to a neural-symbolic cap;
6. later decide whether successful fibres should remain external, write into intermediate layers, or be partially absorbed into the base.

This route makes the scientific comparison cleaner. Every arm uses the same backbone. Differences can be attributed to the residual architecture, state solver, credit rule, causal routing, and symbolic interface rather than to an entirely different from-scratch training history.

Design rule. Do not require the new learning rule to relearn language, discover a causal basis, learn long-range credit, preserve old behavior, and create a symbolic interface simultaneously. Preserve the competent substrate and make the structured residual solve the smaller problem first.

4 Architecture: a frozen backbone with a causal-fibre sidecar

4.1 Frozen computation and selected observation sites

Let the frozen backbone have L blocks:

$$h_0^0 = \text{Embed}(x), \quad (17)$$

$$h_\ell^0 = F_\ell^0(h_{\ell-1}^0), \quad \ell = 1, \dots, L, \quad (18)$$

$$a^0 = U_0(h_L^0). \quad (19)$$

The superscript 0 emphasizes that these states belong to the unmodified base. Choose a set of observation sites

$$\mathcal{S} = \{\ell_1, \dots, \ell_m\}. \quad (20)$$

The first terminal-sidecar experiments cache h_ℓ^0 once and perform all iterative settlement in the small residual state. This avoids rerunning the full Transformer at every PC step.

4.2 Canonical fibres and local charts

A causal fibre is not defined as a layer. It is a canonical latent or parameter subspace that can have local attachments at several layers or modules. Let

$$z_q \in U_q \cong \mathbb{R}^{d_q}, \quad q = 1, \dots, Q. \quad (21)$$

At frozen site ℓ , a read map produces a fibre-specific observation

$$r_{\ell q} = A_{\ell q} \text{LN}(h_\ell^0). \quad (22)$$

The local chart is modeled as

$$r_{\ell q} \approx T_{\ell q} z_q, \quad (23)$$

with registration error

$$\epsilon_{\ell q}^{\text{reg}} = r_{\ell q} - T_{\ell q} z_q. \quad (24)$$

This allows one mechanism to have different coordinates at different depths while retaining one canonical identity.

An amortized predictor initializes the canonical state:

$$\mu_q = f_{\psi_q}(r_{\ell_1 q}, \dots, r_{\ell_m q}, c), \quad z_q^{(0)} = \mu_q. \quad (25)$$

Here c may encode a task, environment, prompt regime, or inferred context.

4.3 Activation routing and plasticity routing

The forward contribution of fibre q is controlled by an activation gate

$$a_q(x, c) \in [0, 1]. \quad (26)$$

Persistent learning is controlled separately by a write-permission or plasticity gate

$$w_q(x, c) \in [0, 1]. \quad (27)$$

A shared fibre can therefore be read in many contexts while being writable only when evidence is judged novel. Conflating these two gates makes it difficult to implement read-mostly shared memory.

4.4 Terminal write path

The safest initial write is terminal:

$$\delta h_L = \gamma \sum_{q=1}^Q a_q B_q z_q, \quad (28)$$

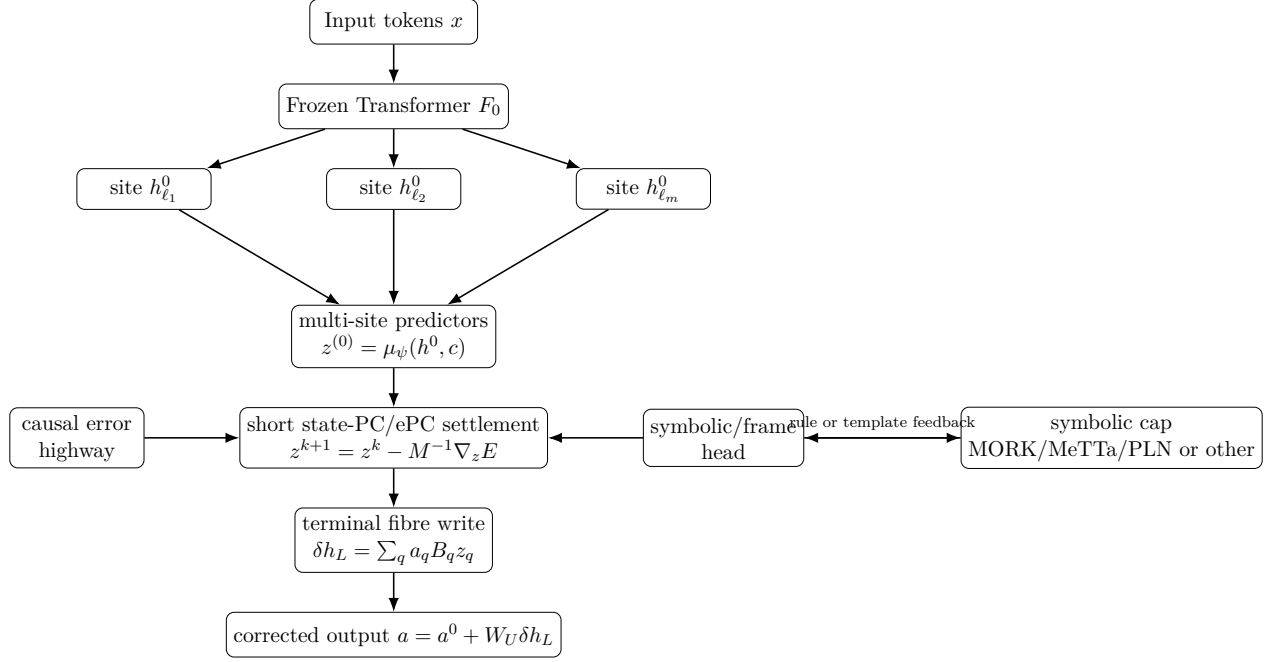


Figure 5: Reference terminal-sidecar architecture. The base remains frozen. Causal fibres live in a small canonical residual state, while routing, coordinate transport, and symbolic registration are explicit.

with corrected logits

$$a = a^0 + W_U \delta h_L. \quad (29)$$

A direct logit writer is also possible:

$$a = a^0 + \sum_q a_q D_q z_q. \quad (30)$$

The residual scale γ can be initialized at zero and warmed up. Setting $\gamma = 0$ recovers the frozen base exactly.

4.5 Intermediate writes and later absorption

After the terminal sidecar works, one can introduce an intermediate write at site ℓ^* :

$$\tilde{h}_{\ell^*} = h_{\ell^*}^0 + \sum_q a_{\ell^*q} B_{\ell^*q} T_{\ell^*q} z_q, \quad (31)$$

followed by frozen suffix recomputation

$$\tilde{h}_L = F_{L:\ell^*+1}^0(\tilde{h}_{\ell^*}). \quad (32)$$

This is more expressive but more invasive and more expensive during settlement. The recommended progression is terminal logits, terminal hidden state, one late intermediate write, several sparse intermediate writes, and only then partial absorption into base parameters.

5 Residual predictive coding and partial distillation

5.1 External objective

Let p_T be the teacher distribution, p_R the frozen-base-plus-residual distribution, and p_0 the frozen base distribution. A generic external objective is

$$\mathcal{L}_{\text{ext}} = \lambda_{\text{CE}} \text{CE}(y, p_R) + \lambda_{\text{KD}} \tau^2 \text{KL}(p_T^\tau \| p_R^\tau) \quad (33)$$

$$+ \lambda_{\text{anchor}} \mathbf{1}_{x \in \mathcal{A}} \text{KL}(p_0 \| p_R). \quad (34)$$

The anchor set $\mathcal{A}$ contains behavior the residual should preserve. This makes “do not damage the competent base” an explicit constraint rather than a hope.

A useful teacher-gap closure metric is

$$\Gamma_L = \frac{L_0 - L_R}{L_0 - L_T}, \quad (35)$$

for a loss where lower is better. For accuracy,

$$\Gamma_A = \frac{A_R - A_0}{A_T - A_0}. \quad (36)$$

5.2 Residual free energy

A practical residual free energy is

$$E(z) = \mathcal{L}_{\text{ext}}(z) + \frac{1}{2} \sum_q \|z_q - \mu_q\|_{\Pi_q}^2 + \frac{\lambda_{\text{abs}}}{2} \sum_q \|z_q\|^2 \quad (37)$$

$$+ \frac{\lambda_{\text{reg}}}{2} \sum_{\ell, q} \|r_{\ell q} - T_{\ell q} z_q\|_{\Lambda_{\ell q}}^2 + \lambda_{\text{sym}} E_{\text{sym}} + \lambda_{\text{other}} E_{\text{other}}. \quad (38)$$

The predictor prior encourages fast target-free inference. The absolute state term fixes a scaling gauge. Registration terms encourage one canonical fibre to remain coherent across local charts.

The desired monotonicity property is

$$E(z^{k+1}) \leq E(z^k) \quad (39)$$

across settlement iterations. It is *not* a requirement that raw activity decrease monotonically through Transformer depth.

5.3 State-PC update

The simplest state inference is

$$z^{k+1} = z^k - \eta_k \nabla_z E(z^k). \quad (40)$$

Backtracking can reduce η_k until the energy decreases. This is valuable as a baseline because it exposes whether preconditioning is genuinely helpful.

5.4 Damped Gauss–Newton error predictive coding update

Let

$$J = \frac{\partial a}{\partial z} \quad (41)$$

be the residual-state-to-logit Jacobian, and let G_y be the output Fisher or another positive semidefinite loss metric. A residual Gauss–Newton metric is

$$M_k = J_k^\top G_y J_k + \Pi + \lambda_{\text{reg}} T^\top \Lambda T + (\lambda_{\text{abs}} + \lambda_{\text{LM},k}) I. \quad (42)$$

The proposed step is

$$\Delta z_k = -M_k^{-1} \nabla_z E(z^k), \quad z^{k+1} = z^k + \Delta z_k. \quad (43)$$

For small residual states, a dense Cholesky solve is feasible. For larger states, use block metrics, low-rank Fisher structure, K-FAC, or conjugate gradient with matrix-vector products.

A trust ratio compares actual and predicted energy reduction:

$$\rho_k = \frac{E(z_k) - E(z_k + \Delta z_k)}{-\nabla E_k^\top \Delta z_k - \frac{1}{2} \Delta z_k^\top M_k \Delta z_k}. \quad (44)$$

Increase damping when the quadratic model is unreliable and decrease it when the model predicts the reduction accurately.

5.5 Teacher-clamped versus deployable inference

During training, a teacher or target may shape the settled state

$$z^*(x, p_T) = \arg \min_z E_{\text{clamped}}(z; x, p_T). \quad (45)$$

At deployment, the teacher is absent. The system uses

$$z = \mu_\psi(h^0, c) \quad (46)$$

or optional free settlement under internal and symbolic constraints only.

The predictor should therefore receive both a state-matching and a direct deployment loss:

$$\mathcal{L}_{\text{amort}} = \lambda_z \|\mu_\psi - \text{sg}(z^*)\|_\Pi^2 + \lambda_{\text{direct}} \mathcal{L}_{\text{ext}}(z = \mu_\psi). \quad (47)$$

The amortization gap

$$\Delta_{\text{amort}} = L_{\text{predictor}} - L_{\text{clamped}} \quad (48)$$

is a primary deployment metric.

5.6 Gauge fixing

The parameterization often admits

$$z_q \mapsto c z_q, \quad B_q \mapsto B_q / c. \quad (49)$$

Raw latent rank and variance can therefore be misleading. Practical gauge controls include

$$\|B_q\|_F = 1, \quad \text{Cov}(z_q) \approx I, \quad \lambda_{\text{abs}} \|z_q\|^2. \quad (50)$$

Diagnostics should report the function-space contribution $B_q z_q$, not only $\|z_q\|$.

Interpretive rule. Teacher-clamped settlement is a credit and capacity diagnostic. Predictor-only behavior is the deployment result. A large clamped advantage with a weak predictor means the architecture can express the correction but has not learned to infer it from target-free inputs.

6 Exact credit, plasticity gating, and optional causal error highways

The conservative default is exact residual autograd combined with responsibility-conditioned persistent plasticity. The highway material in this section is retained because it is a useful optional credit model, not because a small sidecar needs to replace reverse-mode differentiation immediately.

Let $g_q^{\text{oracle}} = \nabla_{\phi_q} L$ be exact credit and $r_{cq} \in [0, 1]$ a context responsibility. The first causal-learning intervention is

$$\tilde{g}_q = r_{cq}^p g_q^{\text{oracle}}. \quad (51)$$

This isolates the continual-learning question: can support estimates protect unrelated modules without corrupting task credit? Only after this baseline succeeds should a learned message $\hat{g}_q^{\text{hw}}$ be considered.

6.1 Why a highway is a separate architectural knob

A metric change can make settlement faster without deciding which mechanism should receive the error. A highway introduces direct long-range reach. A causal highway adds addressability: it sends only the part of an error attributed to a mechanism and only to fibres that can repair that mechanism.

The central decomposition is

$$e_o = e_{\text{shared}} + e_{\text{local}} + e_{\text{nuisance}}, \quad (52)$$

where e_o is a high-level teacher, task, or prediction error. A learned encoder may produce

$$(e_1, \dots, e_J) = C_\chi(e_o, h, c), \quad (53)$$

with mechanism-indexed components e_j .

6.2 Counterfactual factor errors

When controlled interventions are available, define a factor-specific teacher effect

$$\Delta_j^T = f_T(x^{\text{do}(j)}) - f_T(x), \quad (54)$$

and the student's corresponding effect

$$\Delta_j^R = f_R(x^{\text{do}(j)}) - f_R(x). \quad (55)$$

The causal effect error is

$$e_j = P_j(\Delta_j^R - \Delta_j^T), \quad (56)$$

where P_j selects relevant output positions or coordinates. Minimal pairs provide unusually clean supervision for this decomposition.

6.3 Responsibility, novelty, and transport

The highway message to fibre q is

$$m_q = \sum_{j=1}^J r_{qj} T_{q \leftarrow j} N_j(e_j). \quad (57)$$

The terms have different meanings:

- r_{qj} estimates whether a perturbation or update to fibre q can reduce error component j ;
- N_j removes the component already represented by stable shared memory;
- $T_{q \leftarrow j}$ expresses the mechanism-specific error in fibre q 's canonical coordinates;
- a separate write-permission gate decides whether the fibre may change persistently.

A signed repair score can be estimated by

$$R_{qj} = \left[- \left\langle e_j, \hat{J}_{jq} d_q \right\rangle \right]_+, \quad (58)$$

where $\hat{J}_{jq}$ predicts the effect of a candidate fibre update d_q on mechanism output j . A soft responsibility distribution is

$$r_{qj} = \frac{\exp((\log(R_{qj} + \epsilon) - c_q - u_{qj})/T_r)}{\sum_{q'} \exp((\log(R_{q'j} + \epsilon) - c_{q'} - u_{q'j})/T_r)}, \quad (59)$$

with write cost c_q and uncertainty penalty u_{qj} .

6.4 Metric-aware transport

Even a causally correct residual can be harmful in the wrong coordinates. Let G_j be the metric of error component j and M_q the fibre metric. A whiten-transport-unwhiten parameterization is

$$\bar{e}_j = G_j^{-1/2} e_j, \quad (60)$$

$$m_{qj} = M_q^{1/2} \bar{T}_{qj} \bar{e}_j. \quad (61)$$

The natural state step uses $M_q^{-1} m_{qj}$. Constraining $\bar{T}_{qj}$ near a partial isometry prevents arbitrary amplification.

Two transport regularizers are especially important:

$$\mathcal{L}_{\text{path}} = \|T_{a \leftarrow b} T_{b \leftarrow o} - T_{a \leftarrow d} T_{d \leftarrow o}\|_F^2, \quad (62)$$

$$\mathcal{L}_{\text{metric}} = \left\| T^\top G_{\text{local}} T - G_{\text{canonical}} \right\|_F^2. \quad (63)$$

Intervention alignment can additionally use

$$\mathcal{L}_{\text{int-trans}} = \left\| T_{\ell q} \Delta z_q - \Delta h_\ell^{\text{do}(q)} \right\|^2. \quad (64)$$

6.5 Credit-only versus state-coupling highways

A *credit-only* highway modifies latent or parameter updates but is not a new dependency in the forward generative model. This is the recommended first condition. A *state-coupling* highway creates an actual long-range forward or recurrent connection. It may improve representation and transfer, but it changes fixed points and can introduce new entanglement.

6.6 Oracle-blended credit distillation

The small residual permits exact autograd credit as a training oracle. Let g_q^{oracle} be the exact task-related residual gradient and m_q the learned highway message. Use

$$g_q^{\text{used}} = \alpha g_q^{\text{oracle}} + (1 - \alpha)m_q + g_q^{\text{local}}. \quad (65)$$

Here g_q^{local} contains predictor, gauge, damping, and local registration forces. It should not be routed through the causal highway.

Train the highway against the whitened oracle:

$$\mathcal{L}_{\text{credit}} = \sum_q \left\| M_q^{-1/2} (m_q - \text{sg}(g_q^{\text{oracle}})) \right\|^2 + \lambda_{\cos} \left[1 - \cos(m_q, g_q^{\text{oracle}}) \right]. \quad (66)$$

Reduce α only after held-out audits show high metric-whitened cosine, low off-support mass, good one-step energy recovery, stable transport conditioning, and no deployment degradation.

Causal credit rule. Broadcast causal responsibility, not raw error. But do not infer from a low task loss that responsibility was routed correctly. Measure the support and the intervention effect of the update itself.

7 Continual learning, shared causes, and commutators

7.1 Context-local supports

Let the trainable residual parameters be grouped into modules

$$\phi = (\phi_1, \dots, \phi_Q). \quad (67)$$

Context c has intended support S_c . Define outside-support gradient mass

$$\epsilon_g(c) = \frac{\sum_{q \notin S_c} \left\| \nabla_{\phi_q} L_c \right\|}{\sum_q \left\| \nabla_{\phi_q} L_c \right\| + \epsilon}. \quad (68)$$

Define cross-support Hessian coupling, schematically,

$$\epsilon_h(c) = \max_{q \in S_c, r \notin S_c} \left\| \nabla_{\phi_q \phi_r}^2 L_c \right\|. \quad (69)$$

Causal gates reduce ϵ_g ; modular architecture, clarity, and transport alignment reduce ϵ_h .

7.2 Update commutators

Let

$$X_c(\phi) = -\nabla L_c(\phi) \quad (70)$$

be the context gradient field. For gradient flows,

$$[X_c, X_d] = -\nabla^2 L_d \nabla L_c + \nabla^2 L_c \nabla L_d. \quad (71)$$

When supports are disjoint and outside-support gradients and cross-support Hessians are small,

$$\|[X_c, X_d]\| \lesssim C(\epsilon_h G_{\max} + \epsilon_g H_{\max}), \quad (72)$$

plus terms arising from genuine support overlap.

For small steps h_c, h_d , the difference between update orders is second order:

$$U_d^{h_d} U_c^{h_c} \phi - U_c^{h_c} U_d^{h_d} \phi = \frac{1}{2} h_c h_d [X_c, X_d](\phi) + O(h^3). \quad (73)$$

This motivates direct function-space order tests:

$$\phi_{cd} = U_d U_c \phi, \quad (74)$$

$$\phi_{dc} = U_c U_d \phi, \quad (75)$$

$$D_f(c, d) = \mathbb{E}_x \text{SKL}(p_{\phi_{cd}}(\cdot | x), p_{\phi_{dc}}(\cdot | x)). \quad (76)$$

7.3 Metric mismatch versus path dependence

Forgetting can contain two qualitatively different contributions.

1. **Modewise multiplicity mismatch.** A schedule gives a fixed evidence mode too much or too little effective weight. Natural gradient, precision matching, replay weighting, or schedule correction can help.
2. **Path-dependent causal reuse.** Shared evidence is re-encoded and transported through noncommuting representations. Causal residualization, modularity, clarity, and curvature-aware routing address this channel.

A metric improvement does not automatically flatten a curved task connection, and modularity does not automatically calibrate the evidence schedule. Both should be instrumented.

7.4 Shared causal memory

A genuine common cause should be represented once and read by many contexts. Let K_j span assimilated innovation directions for mechanism j . The metric projector is

$$P_j^{\text{known}} = K_j (K_j^\top G_j K_j + \epsilon I)^{-1} K_j^\top G_j. \quad (77)$$

The new component is

$$e_j^{\text{nov}} = (I - P_j^{\text{known}}) e_j. \quad (78)$$

A novelty-gated core write is

$$w_j^{\text{core}} = \sigma \left(\frac{\|e_j^{\text{nov}}\|_{G_j} - \tau_j}{T_{\text{nov}}} \right). \quad (79)$$

Each mature fibre can have

$$\phi_q = \phi_q^{\text{core}} \oplus \phi_q^{\text{shell}}. \quad (80)$$

The core is stable, reusable, and write-gated. The shell is context-local and plastic. Stable shell structure can later be promoted.

7.5 Mediators rather than forced orthogonality

Some mechanisms interact genuinely. If the residual target has sparse higher-order components, create mediator fibres rather than forcing all singleton fibres to be orthogonal. A Fisher synergy estimate is

$$\text{Syn}(q, r) = \delta h_q^\top G \delta h_r. \quad (81)$$

Persistent, context-specific, nonredundant synergy is evidence for a mediator. The desired architecture is sparse interpretable coupling, not zero coupling.

8 When should causal fibres emerge?

8.1 The operator view

The central mathematical idea is that a causal decomposition should be visible across a family of context-conditioned operators, not only in one activation covariance. Let

$$H_c \succeq 0 \quad (82)$$

be a Fisher, Gauss–Newton, Hessian, empirical-Fisher, linearized-update, or update-covariance operator for context c in the compact residual-state or residual-parameter space.

Let U_q be candidate fibre subspaces with orthogonal projectors P_q . Exact fibre locality means

$$P_q H_c P_r = 0, \quad q \neq r, \quad (83)$$

for all contexts. A joint block-diagonalization objective is

$$\Phi(P_1, \dots, P_Q) = \sum_c \sum_{q < r} \|P_q H_c P_r\|_F^2. \quad (84)$$

This objective asks for one decomposition that makes the entire operator family block diagonal.

8.2 Common blocks are sufficient, not universal

The common-block model is mathematically clean and valuable when supported, but a Transformer may use superposition. Let $S = D^\top$ be a synthesis map for an overcomplete dictionary. A minimum-norm coordinate representation of a hidden-space operator is

$$K_c = S^+ H_c S, \quad (85)$$

where S^+ is the Moore–Penrose pseudoinverse. The reconstruction audit is

$$\epsilon_{\text{op},c} = \frac{\|H_c - SK_c S^+\|_F}{\|H_c\|_F + \epsilon}. \quad (86)$$

One may then measure coordinate sparsity and group-off-support mass. These coordinates are generally non-unique and do not prove causal identification; they provide a rival structural chart in which sparse mechanism groups may be more natural than orthogonal subspaces.

For context-dependent charts, write

$$H_c \approx T_c D A_c D^\top T_c^\top + R_c. \quad (87)$$

The relevant questions become whether T_c is low complexity, whether its paths are consistent, whether its holonomy is concentrated in genuine shared mechanisms, and whether the canonical code supports stable interventions or symbolic registration.

8.3 Exact joint-spectrum identification

Theorem 8.1 (Exact causal-fibre identification). *Suppose*

$$H_c = \sum_{q=1}^Q \lambda_{cq} P_q \quad (88)$$

for every context c , where the P_q are mutually orthogonal projectors. Define the context signature

$$\boldsymbol{\lambda}_q = (\lambda_{1q}, \dots, \lambda_{Cq}). \quad (89)$$

If

$$\boldsymbol{\lambda}_q \neq \boldsymbol{\lambda}_r \quad q \neq r, \quad (90)$$

then the H_c commute, the $U_q = \text{im } P_q$ are their joint eigenspaces, and the decomposition is unique up to permutation of blocks and orthogonal changes of coordinates inside each block. Among decompositions with the correct block dimensions, the zero minima of Φ are exactly these fibre subspaces.

Proof sketch. Every H_c is a linear combination of the same projectors, so the family commutes. The simultaneous spectral theorem yields a common eigenspace decomposition. Vectors in U_q have joint eigenvalue signature $\boldsymbol{\lambda}_q$. Distinct signatures prohibit mixing between different U_q . A zero of Φ gives projectors commuting with the full family, so each candidate block is a sum of joint eigenspaces; the rank constraints isolate the stated blocks. $\square$

The key identifiability requirement is therefore not merely low interference. Different mechanisms must be modulated differently by at least one context or intervention.

8.4 The two-fibre rotation calculation

Let

$$H_c = \begin{pmatrix} \lambda_{c1} & 0 \\ 0 & \lambda_{c2} \end{pmatrix} \quad (91)$$

in the true basis, and rotate the candidate basis by angle θ . The off-diagonal element is

$$\left(R_\theta^\top H_c R_\theta\right)_{12} = \frac{1}{2}(\lambda_{c2} - \lambda_{c1}) \sin 2\theta. \quad (92)$$

Therefore

$$\Phi(\theta) = \frac{1}{4}\Delta_{12}^2 \sin^2 2\theta, \quad \Delta_{12}^2 = \sum_c (\lambda_{c1} - \lambda_{c2})^2. \quad (93)$$

When $\Delta_{12} > 0$, the true basis and its permutation are strict minima. Near the true basis,

$$\Phi(\theta) = \Delta_{12}^2 \theta^2 + O(\theta^4). \quad (94)$$

The signature gap is the restoring curvature. If $\Delta_{12} = 0$, every rotation is equally valid: no training algorithm can identify two distinct fibres from the available contexts.

8.5 Approximate blocks and an emergence ratio

For

$$H_c = \begin{pmatrix} a_c & b_c \\ b_c & d_c \end{pmatrix}, \quad (95)$$

set $\delta_c = (d_c - a_c)/2$ and define

$$A = \sum_c \delta_c^2, \quad B = \sum_c b_c^2, \quad C = \sum_c \delta_c b_c. \quad (96)$$

The optimum satisfies

$$\tan 4\theta^* = -\frac{2C}{A - B}. \quad (97)$$

When $A \gg B$,

$$|\theta^*| \lesssim \frac{1}{2} \sqrt{\frac{B}{A}}. \quad (98)$$

This suggests the pairwise fibre-emergence ratio

$$\Gamma_{qr} = \frac{\text{context-signature separation}}{\text{cross-fibre coupling} + \text{operator-estimation noise}}. \quad (99)$$

A concrete multidimensional form is

$$\Gamma_{qr} = \frac{\sum_c \|H_c^{(q)} - H_c^{(r)}\|_F^2}{4 \sum_c \|P_q H_c P_r\|_F^2 + \sigma_{\text{est}}^2}. \quad (100)$$

Large Γ_{qr} predicts stable separation; values near or below one predict mixed or seed-sensitive fibres.

8.6 Perturbation bound

Suppose

$$\hat{H}_c = H_c^0 + E_c, \quad H_c^0 = \sum_q \lambda_{cq} P_q. \quad (101)$$

For coefficients α_c , form

$$H_\alpha^0 = \sum_c \alpha_c H_c^0, \quad (102)$$

with block eigenvalues

$$\mu_q = \sum_c \alpha_c \lambda_{cq}. \quad (103)$$

Let

$$\delta_\alpha = \min_{q \neq r} |\mu_q - \mu_r|. \quad (104)$$

A Davis–Kahan type estimate gives, schematically,

$$\left\| \sin \Theta(\hat{U}_q, U_q) \right\| \leq \frac{2 \left\| \sum_c \alpha_c E_c \right\|}{\delta_\alpha}, \quad (105)$$

when the perturbation is below the gap. Thus operator error must be small relative to context-signature separation.

8.7 The operator algebra and multiplicity

Let

$$\mathcal{A} = \text{alg}^* \{H_c : c \in \mathcal{C}\} \quad (106)$$

be the self-adjoint algebra generated by the operator family. Causal fibres correspond to central invariant blocks of $\mathcal{A}$. The favorable case is multiplicity free:

$$\mathcal{A} \cong \bigoplus_q M_{d_q}(\mathbb{R}). \quad (107)$$

The underidentified case contains repeated copies:

$$\mathcal{A} \cong M_d(\mathbb{R}) \otimes I_m. \quad (108)$$

Any orthogonal mixing in the multiplicity space is equivalent. Repeated identical columns or fibres therefore require extra identifying contexts, semantic anchors, or controlled detuning; optimization alone cannot choose a uniquely causal split.

8.8 Symmetry breaking

If Q fibres have identical architecture, identical initialization, a permutation-invariant objective, and deterministic gradient updates, then

$$\theta_1(0) = \dots = \theta_Q(0) \quad (109)$$

implies

$$\theta_1(t) = \dots = \theta_Q(t) \quad (110)$$

for all steps. Independent initialization is necessary, and small differences in precision, regularization, symbolic anchoring, or initial context affinity can improve identifiability. Excessive detuning, however, can bias the solution away from data-supported factors.

8.9 Existence before identification

Let the teacher-minus-base residual depend on candidate causal variables $v = (v_1, \dots, v_K)$ and admit a functional decomposition

$$r(v) = r_\emptyset + \sum_i r_i(v_i) + \sum_{i < j} r_{ij}(v_i, v_j) + \dots \quad (111)$$

Define interaction mass

$$\eta_{\text{int}} = \frac{\sum_{|S| \geq 2} \|r_S\|_G^2}{\sum_{|S| \geq 1} \|r_S\|_G^2}. \quad (112)$$

Small η_{int} supports singleton fibres. Sparse concentrated interactions support singleton fibres plus mediator fibres. Broad high-order interaction mass is evidence against a small clean fibre model.

8.10 Observability and controllability

For mechanism q , define best read-side prediction error

$$\epsilon_{\text{obs},q}^2 = \inf_{\mu_q} \mathbb{E} \|z_q - \mu_q(h)\|^2. \quad (113)$$

A plain backprop residual using exactly the proposed reads and writes is a mandatory control. If it cannot close the teacher gap, failure of PC or routing is not informative.

Let

$$J_q = \frac{\partial y}{\partial z_q} \quad (114)$$

be the write Jacobian. Selective controllability requires nonzero gain on the intended output subspace and low crosstalk:

$$\sigma_{\min}(P_{Y_q} J_q) \geq \omega_q > 0, \quad (115)$$

$$\chi_q = \frac{\sum_{r \neq q} \|P_{Y_r} J_q\|_F^2}{\|P_{Y_q} J_q\|_F^2 + \epsilon} \ll 1. \quad (116)$$

8.11 Fingerprint conditioning

Let f_q contain the measured effects of intervening on fibre q across probe contexts, and collect

$$F = [f_1 \ \dots \ f_Q]. \quad (117)$$

For an s -sparse responsibility vector, stable recovery requires a restricted singular value

$$\kappa_{2s}(F) = \inf_{v \neq 0, \|v\|_0 \leq 2s} \frac{\|Fv\|_2}{\|v\|_2} > 0. \quad (118)$$

Nearly identical fingerprints make learned routing unstable even if distinct fibres exist architecturally.

8.12 Transport flatness and fibre bundles

Let $T_e^{(q)}$ be transports along edges of a graph of local representations. A globally consistent canonical fibre requires

$$\prod_{e \in \gamma} T_e^{(q)} = I \quad (119)$$

for every cycle γ . Approximate cycle defect is transport holonomy. If it is small, one canonical coordinate system is reasonable. If it is large but structured, use local charts and explicit mediators. If it is broad and incoherent, a global fibre may not be stable.

8.13 Local emergence theorem template

Theorem 8.2 (Local approximate emergence, schematic). *Assume the residual target is approximately factorable; the context operators have the form*

$$H_c = \sum_q \lambda_{cq} P_q + E_c; \quad (120)$$

context signatures have minimum gap $\Delta_{\min} > 0$; the reads are observable; writes are selectively controllable; the fingerprint matrix is conditioned; transport holonomy is small; the scale gauge is fixed; and optimization reaches the relevant basin. Then recovered fibre subspaces satisfy a local bound of the form

$$\max_q \left\| \sin \Theta(\hat{U}_q, U_q) \right\| \lesssim \frac{\epsilon_{\text{fact}} + \epsilon_{\text{cross}} + \epsilon_{\text{est}} + \epsilon_{\text{hol}}}{\Delta_{\min}}, \quad (121)$$

while sparse routing error scales inversely with fingerprint conditioning. The resulting outside-support gradient and cross-support Hessian errors control update commutators as in the causal-continual-learning bound.

Remark 8.3. The exact joint-spectrum and perturbation statements are standard finite-dimensional local results. The complete nonlinear learning claim is a theorem template rather than a global convergence theorem. It assumes that the architecture and optimizer enter a basin in which the local operator model is informative.

8.14 Practical feasibility checklist

Precondition	Diagnostic	Failure when absent
Residual factorability	interaction mass or sparse operator dictionary fit	dense entanglement; many mediators required
Context separation	signature gap $\Delta_{\min}$	arbitrary rotations or merged fibres
Read observability	BP residual on identical reads	deployment predictor cannot infer the mechanism
Write controllability	intended gain and crosstalk χ_q	effective correction through unintended outputs
Router identifiability	restricted singular value of fingerprints	unstable responsibility estimates
Transport consistency	path and cycle defects	same mechanism means different things across sites

Multiplicity-free structure	commutant-centre diagnostics	repeated copies are not uniquely identifiable
Stable optimization	gauge, SNR, soft restarts	collapse, duplication, or winner-take-all routing

9 A practical causal-fibre emergence framework

9.1 The framework as online architecture learning

Causal-fibre learning is best treated as an online architecture-learning process rather than one extra regularizer. The system repeatedly answers four questions:

1. Is a fibre decomposition supported by the data and interfaces?
2. Which fibre is responsible for each error component?
3. How should that error be translated into each receiving coordinate system?
4. When should fibres split, merge, become shared, freeze, or disappear?

A candidate architectural slot should not be called a causal fibre immediately. A useful lifecycle is:

Status	Meaning
Candidate	allocated slot with no causal claim
Functional	ablation or perturbation has a distinct effect
Causal	intervention effect is selective and stable across relevant contexts
Consolidated	function and fingerprint remain stable over time and curriculum order
Shared	many contexts read one mechanism, with controlled writes
Mediator	explicit representation of a genuine interaction between other fibres
Retired	no unique value or redundant with another fibre

9.2 Feasibility vector

Before strong modularity pressure, estimate

$$\mathcal{Q}_{\text{fibre}} = (Q_{\text{fact}}, Q_{\text{sep}}, Q_{\text{obs}}, Q_{\text{ctrl}}, Q_{\text{route}}, Q_{\text{transport}}, Q_{\text{sym}}). \quad (122)$$

This vector distinguishes green, amber, and red regimes.

Green.

Factorability, context separation, observability, controllability, fingerprint conditioning, and transport consistency are all adequate. Explicit fibre learning is appropriate.

Amber.

Singleton factors exist but interactions, near-degeneracies, or local transport differences are significant. Use soft routing, mediator fibres, and weak structural penalties.

Red.

Contexts do not separate mechanisms, reads omit necessary information, writes are highly entangled, or interaction structure is dense. Do not force a causal-fibre interpretation.

9.3 Training objective

A generic objective is

$$\mathcal{L}_{\text{CFE}} = \mathcal{L}_{\text{task}} + \lambda_{\text{amort}} \mathcal{L}_{\text{amort}} + \lambda_{\text{block}} \mathcal{L}_{\text{block}} + \lambda_{\text{resp}} \mathcal{L}_{\text{resp}} \quad (123)$$

$$+ \lambda_{\text{leak}} \mathcal{L}_{\text{leak}} + \lambda_{\text{transport}} \mathcal{L}_{\text{transport}} + \lambda_{\text{duplicate}} \mathcal{L}_{\text{duplicate}} \quad (124)$$

$$+ \lambda_{\text{budget}} \mathcal{L}_{\text{budget}} + \lambda_{\text{gauge}} \mathcal{L}_{\text{gauge}}. \quad (125)$$

The terms should be staged rather than activated at full strength from the beginning.

The block term is

$$\mathcal{L}_{\text{block}} = \sum_c \sum_{q \neq r} w_{cqr} \left\| U_q^\top \hat{H}_c U_r \right\|_F^2, \quad (126)$$

with $w_{cqr} = 0$ for approved shared or mediator couplings.

Measured repair effects give a routing target:

$$I_{cq} = -\frac{L_c(z + \epsilon P_q d_q) - L_c(z)}{\epsilon}, \quad \tilde{r}_{cq} = \frac{[I_{cq}]_+}{\sum_j [I_{cj}]_+ + \epsilon}. \quad (127)$$

Then

$$\mathcal{L}_{\text{resp}} = \sum_c \text{KL}(\tilde{r}_c \| r_\omega(c, h, e)). \quad (128)$$

An off-support penalty is

$$\mathcal{L}_{\text{leak}} = \sum_{c,q} (1 - \text{sg}(r_{cq})) \left\| \nabla_{\phi_q} L_c \right\|^2. \quad (129)$$

A conservative duplicate penalty requires agreement in several spaces:

$$\mathcal{L}_{\text{duplicate}} = \sum_{q < r} s_{\text{fp}}(q, r) s_{\text{read}}(q, r) s_{\text{write}}(q, r). \quad (130)$$

Correlated activation alone is not sufficient evidence of duplication.

9.4 Soft routing before hard routing

Use soft top- k , sparsemax, or entmax initially. A budget term can be

$$\mathcal{L}_{\text{budget}} = \sum_i \left(\sum_q a_q(x_i, c_i) - k \right)^2 + \beta \sum_i H(a(x_i, c_i)). \quad (131)$$

Hard top-1 routing should be delayed until transport and intervention audits show that one fibre or a small coalition carries the correct mechanism-specific update.

9.5 Causal Fibre EM

The complete learning loop is EM-like: infer latent states and responsibility, update fibre content, update structure, and choose informative future contexts.

```
initialize fibre predictors, writers, soft router, transports, metrics
initialize reserve fibres and stable fibre identifiers
```

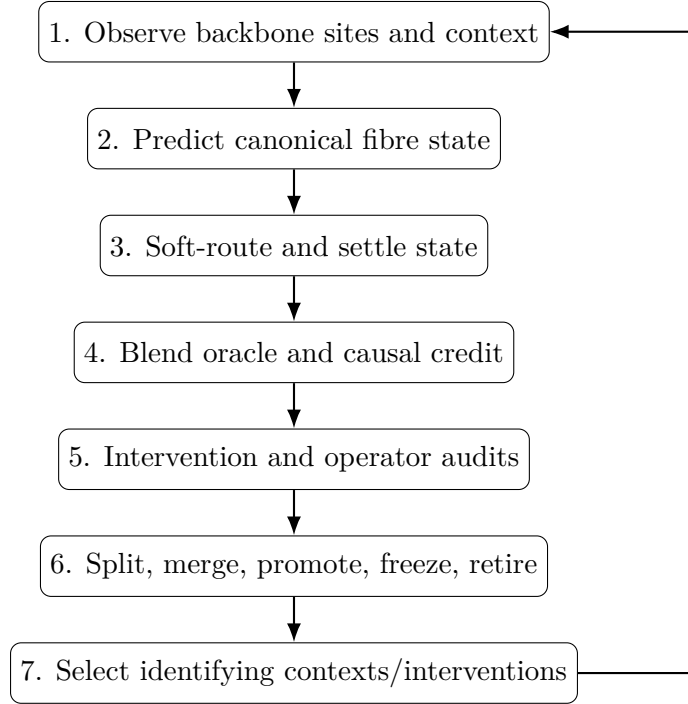


Figure 6: Causal Fibre EM. Fibre emergence is an iterative measurement and architecture-learning process, not a single static loss term.

```

set oracle credit blend alpha close to 1

for minibatch (x, y, context c):
    h = architecture.forward_base(x)
    z0 = predictor(h, c)

    z = z0
    for k in range(K_settle):
        task_error = compute_task_or_teacher_error(z)
        factor_errors = factorize_error(task_error, h, c)
        novelty_errors = novelty_filter(factor_errors)
        responsibility = router(h, c, factor_errors)
        causal_credit = transport(responsibility, novelty_errors)
        z = damped_metric_step(z, local_energy, causal_credit)

    deployment_loss = task_loss(z0)
    amortization_loss = distance(z0, stop_gradient(z))

    oracle_credit = exact_residual_vjp_if_available()
    used_credit = alpha * oracle_credit + (1 - alpha) * causal_credit
    update_residual_parameters(used_credit, local_regularizers)

    periodically:
        run_true_interventions()
        update_causal_fingerprints()
        estimate_context_operators()
        jointly_block_diagonalize()
  
```

```

    update_transport_calibration()
    propose_lifecycle_events()
    choose_more_identifying_contexts()

    anneal alpha and router temperature only when held-out audits pass

```

Listing 1: Architecture-neutral causal fibre training loop.

9.6 Active curriculum design

The curriculum should create identifying variation. Given a current fingerprint matrix F , choose an intervention ι to improve conditioning:

$$\iota^* = \arg \max_{\iota} \left[\log \det(F_{\text{aug}(\iota)}^\top F_{\text{aug}(\iota)} + \epsilon I) - \lambda_{\text{cost}} C(\iota) \right]. \quad (132)$$

A simpler criterion maximizes the smallest signature separation induced by the new context. Minimal counterfactual pairs, selective environment changes, mediator-specific interventions, and deliberately rotated-overlap controls are useful.

9.7 Lifecycle rules

Spawn. Let $e_{\text{unexplained}} = e - \sum_q J_q d_q$. Spawn a reserve fibre when $\|e_{\text{unexplained}}\|_G$ is persistently large and existing fibres have low responsibility.

Split. Split fibre q when its fingerprint is multimodal or its internal operator family admits a stable lower block decomposition:

$$\min_{U_{q1} \oplus U_{q2} = U_q} \sum_c \|P_{q1} H_c P_{q2}\|_F^2 < \tau_{\text{split}}. \quad (133)$$

Merge. Merge fibres only when fingerprints, read subspaces, and write subspaces are all close and held-out loss changes negligibly.

Promote. Move stable, widely reused, low-curvature structure into a shared core with a reduced learning rate and novelty-gated writes.

Retire. Retire a fibre when it has negligible unique ablation value, duplicates another fibre, or lacks a stable role for a prolonged audit interval.

10 Robust joint block diagonalization

10.1 Objective and role

Given symmetric operators $H_1, \dots, H_C \in \mathbb{R}^{n \times n}$ and requested block sizes $d_1, \dots, d_Q$, seek an orthogonal basis

$$Q = [U_1 \mid \dots \mid U_Q] \quad (134)$$

minimizing

$$L_{\text{JBD}}(Q) = \sum_c \sum_{q < r} \left\| U_q^\top H_c U_r \right\|_F^2. \quad (135)$$

The fitted loss measures operator cross-block energy. It does not by itself prove causal semantics; signature gaps, restart stability, intervention fingerprints, and underidentification diagnostics must also be inspected.

10.2 Why the first optimizer failed

The initial single-start implementation optimized a QR-parameterized basis from a near-identity start. It succeeded for simple local rotations but often remained in poor local minima when the true common basis was an arbitrary distant rotation and each block contained internally noncommuting operators. The operator metrics could audit a supplied basis, but automatic discovery was not reliable.

10.3 Robust global-to-local algorithm

The robust algorithm uses several complementary initializers.

10.3.1 Spectral combinations

For random coefficients α_c , form

$$M(\alpha) = \sum_c \alpha_c H_c. \quad (136)$$

Every generic combination shares the true invariant blocks. Let V be its eigenbasis. Build an affinity graph

$$W_{ij} = \left(\sum_c \frac{|(V^\top H_c V)_{ij}|^2}{\|H_c\|_F^2} \right)^{1/2}. \quad (137)$$

Cross-block edges vanish in the exact case. Group eigenvectors under the requested block capacities.

10.3.2 Approximate commutant

Solve

$$\min_X \sum_c \|H_c X - X H_c\|_F^2 \quad (138)$$

over symmetric X . Low-residual commutant modes contain central elements whose eigenspaces expose common invariant blocks. The dimension and centre of the approximate commutant also diagnose repeated-copy underidentification.

10.3.3 Capacity-constrained partitioning

For small n , exact dynamic programming maximizes within-block affinity. Larger dense problems use balanced spectral clustering, Hungarian assignment to enforce capacities, and pair swaps to improve affinity.

10.3.4 Multi-start Riemannian refinement

Spectral, commutant, identity, user-supplied, and random candidates are scored and refined on the orthogonal group. The Euclidean gradient is projected to the tangent space:

$$G_R = G - Q \operatorname{sym}(Q^\top G). \quad (139)$$

A retraction and Armijo backtracking preserve orthogonality and avoid objective-increasing forced steps. Near-tied solutions are resolved by context-signature separation.

10.4 Identifiability diagnostics

A fit can return statuses such as

- `multiplicity_free`;
- `repeated_or_underidentified`;
- `finer_structure_detected`;
- `approximate_or_noisy`.

A zero block loss with a near-zero signature gap and an oversized commutant is not evidence for a unique causal decomposition.

```
from causal_fibres.operators import JointBlockDiagonalizer

solver = JointBlockDiagonalizer(
    block_sizes=[8, 8, 8, 8],
    initialization="hybrid",
    spectral_restarts=8,
    commutant_restarts=4,
    random_restarts=2,
    steps=300,
    random_seed=7,
)
result = solver.fit(context_operators)

print(result.initialization)
print(result.report.to_dict())
for restart in result.restarts:
    print(restart.to_dict())
```

Listing 2: Robust joint block diagonalization in `causal-fibres`.

11 Sandbox evidence

11.1 Evidence ladder

The sandbox work was intentionally staged. Each experiment isolated one failure mode or one proof-of-mechanism rather than attempting a natural-language benchmark immediately.

ID	Question	Main conclusion
S0	Does the original full-network method distribute credit and preserve factors?	No: almost all activity-energy-objective credit was confined to the final block, and factor-relevant rank and selectivity contracted.

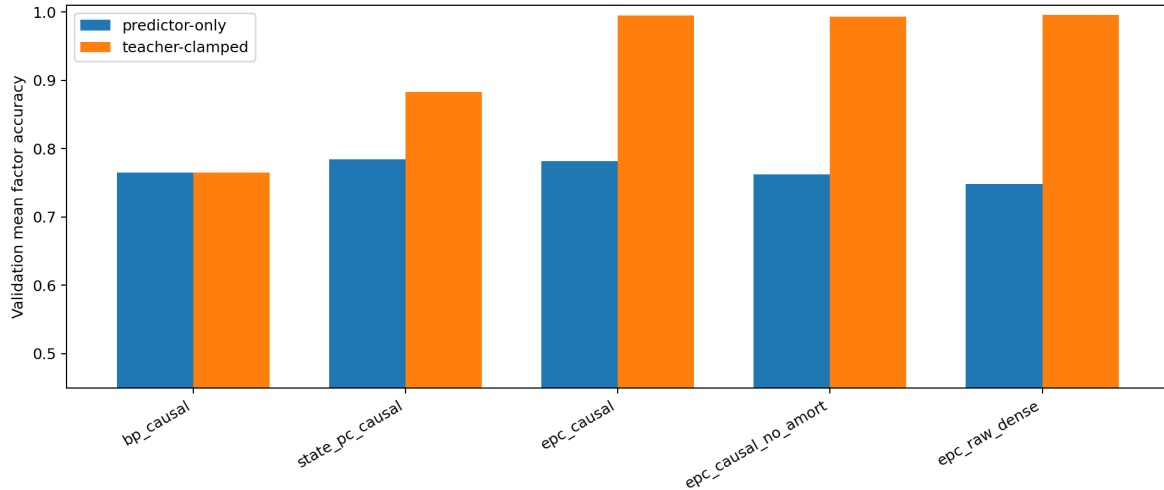


Figure 7: Predictor-only deployment versus teacher-clamped settlement in the first sidecar experiment. The large early gap motivated explicit deployment KD in addition to settled-state regression.

S1	Can a frozen sidecar settle a useful state, and what can go wrong?	Yes in clamped mode, but deployment can lag; latent/write gauge and weak damping can masquerade as collapse or instability.
S2	Does causal routing remain correct under coordinate rotation?	Only with explicit transport calibration. Low energy and good task loss can coexist with badly misallocated causal credit.
S3	Can a deliberately canonical fibre system show early success?	Yes: it recovered most of a teacher gap and gave zero unrelated drift under supplied-support continual adaptation.
S4	Do the software abstractions work together?	Yes in toy tests: distillation, settlement, continual plasticity, intervention fingerprints, symbolic retrieval, and operator audits all ran successfully.
S5	Can arbitrary hidden invariant blocks be recovered robustly?	The robust global-to-local solver recovered exact and noisy common blocks and flagged repeated-copy underidentification.

11.2 S1: first frozen-sidecar sanity test

A small two-block Transformer teacher and a smaller frozen base were used on four generated grammar factors. A four-fibre sidecar read cached states and wrote a terminal correction. At 600 updates, the ePC causal condition reached about 0.995 teacher-clamped accuracy but only about 0.782 predictor-only accuracy. The result exposed a large amortization gap.

At longer training, ordinary BP remained the strongest deployable residual learner in this toy. ePC settlement was useful as an inner credit and inference mechanism but was not demonstrated to be a better outer optimizer. This motivated an exact residual-gradient oracle and gradual credit distillation rather than immediate replacement of backpropagation.

The experiment also revealed a scale-state gauge. One fibre acquired enormous latent variance while its output scale shrank. Raw effective rank appeared near one, but fibre-balanced rank was much healthier.

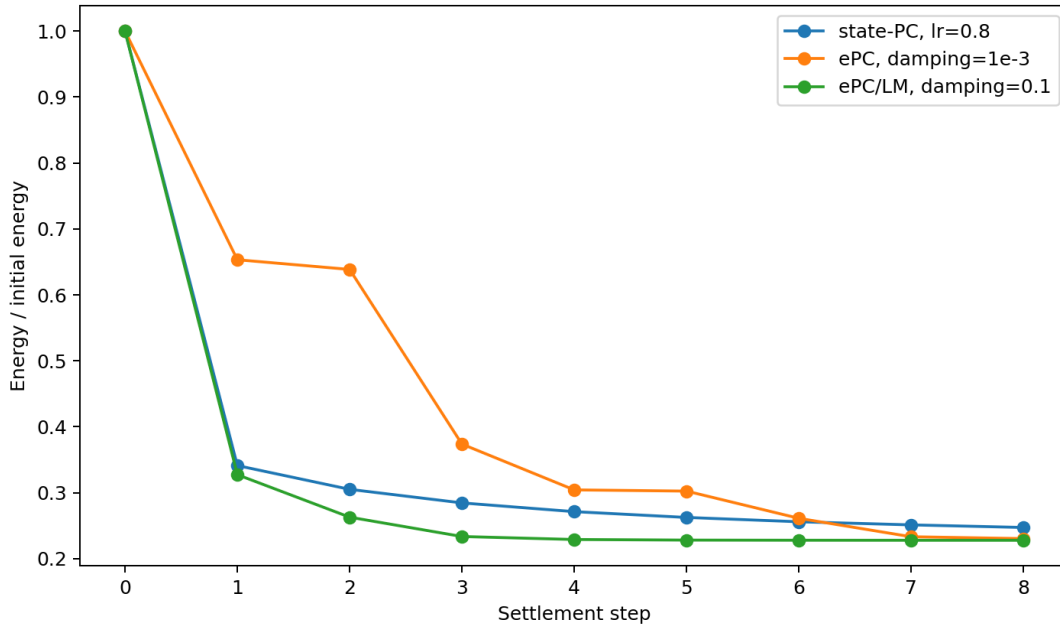


Figure 8: Settlement behavior under different metrics and damping values in the first sandbox. Stronger damping converted a fragile nominal Gauss–Newton step into a reliable trust-region update.

Adding an absolute state prior and fixing or normalizing write scales stabilized the representation.

A damping sweep showed that a nominal Gauss–Newton step with damping 10^{-3} backtracked repeatedly, while damping near 0.1 produced faster monotone reduction. The useful conclusion was not one universal damping value; it was that adaptive Levenberg–Marquardt control is necessary.

A supplied-support continual-learning control showed the expected structural upper bound: strict fibre-local updates produced zero off-support gradients and nearly zero unrelated-output drift, while dense routing changed unrelated behavior.

11.3 S2: rotated-coordinate causal highway

The next experiment held the underlying canonical causal fibres fixed but rotated their local coordinates by 45 or 75 degrees. It compared dense credit, hard raw sparse routing, learned canonical transport, and oracle transport. Three-seed results are summarized below.

Rotation	Credit mechanism	Hard cosine	Off-support mass	Energy fraction
45°	Dense	0.815	0.183	0.989
45°	Raw sparse	0.546	0.521	1.012
45°	Learned transport	0.807	0.066	1.015
45°	Oracle transport	0.837	0.0004	1.011
75°	Dense	0.781	0.234	0.994
75°	Raw sparse	0.628	0.307	0.950
75°	Learned transport	0.551	0.386	0.713
75°	Oracle transport	0.831	0.000	1.050

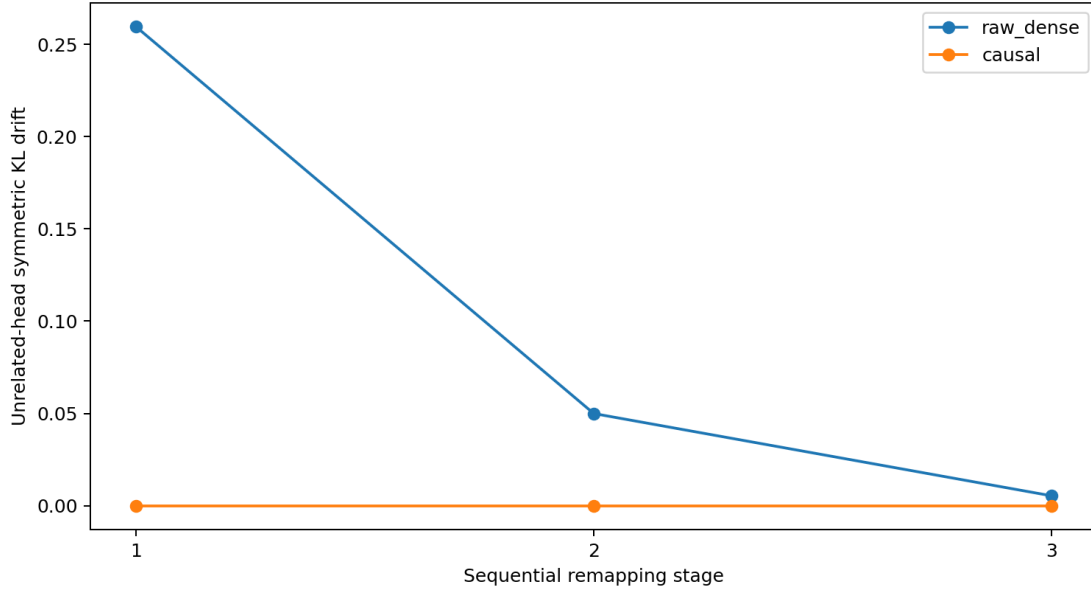


Figure 9: Unrelated-output drift under dense versus structurally local adaptation in the first sandbox. This is a supplied-support control, not a learned-routing result.

The hard raw sparse condition at 45° recovered about all of the one-step energy reduction while placing more than half of its update outside canonical support. This is a direct demonstration that good energy descent does not imply correct causal credit.

The oracle-transport condition succeeded even at 75° , so the architecture itself was adequate. The failure was transport identification. End-to-end credit matching allowed the direction generator and transport matrix to compensate for one another. This motivated separate intervention, path, and metric anchors for transport.

11.4 S3: canonical error fibres

The third sandbox deliberately defined a coordinate system in which early success should be possible. For each binary factor q , the canonical teacher-minus-base log-odds correction was

$$\delta_q(x) = \text{logit } p_{T,q}(x) - \text{logit } p_{0,q}(x). \quad (140)$$

Each fibre predicted one scalar s_q with fixed write

$$(a_{q,0}, a_{q,1}) \mapsto (a_{q,0} - s_q/2, a_{q,1} + s_q/2). \quad (141)$$

Thus $s_q = \delta_q$ exactly reproduces the teacher probability for factor q . The write basis fixes the gauge and gives each fibre a precise meaning.

Across three seeds and 1,600 updates:

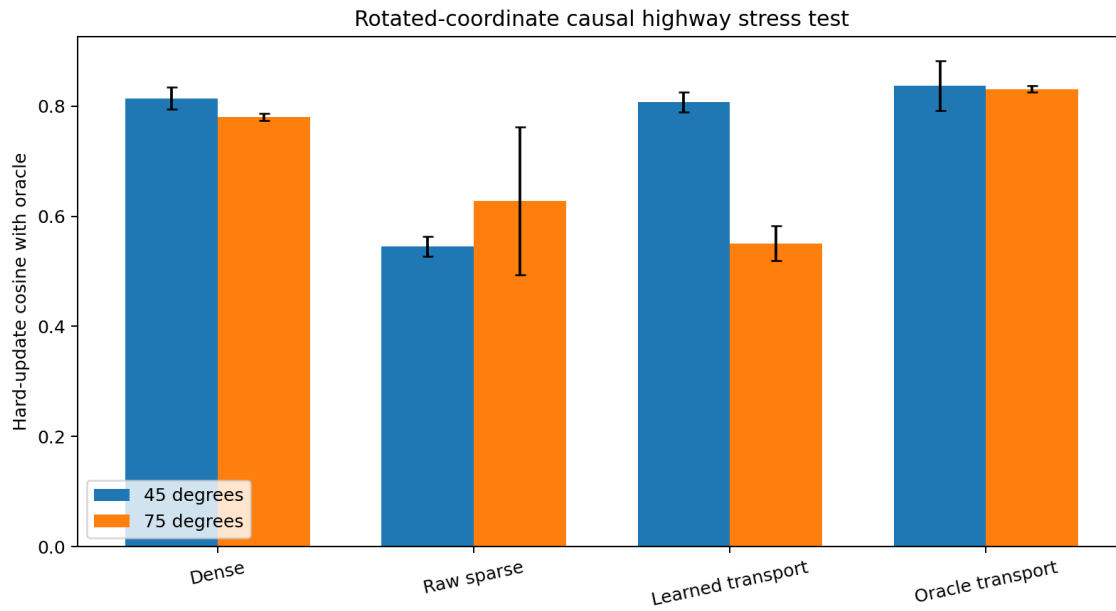


Figure 10: Hard-update cosine with the exact residual oracle under coordinate rotation. Learned transport works at moderate rotation but becomes underidentified at severe rotation unless separately anchored.

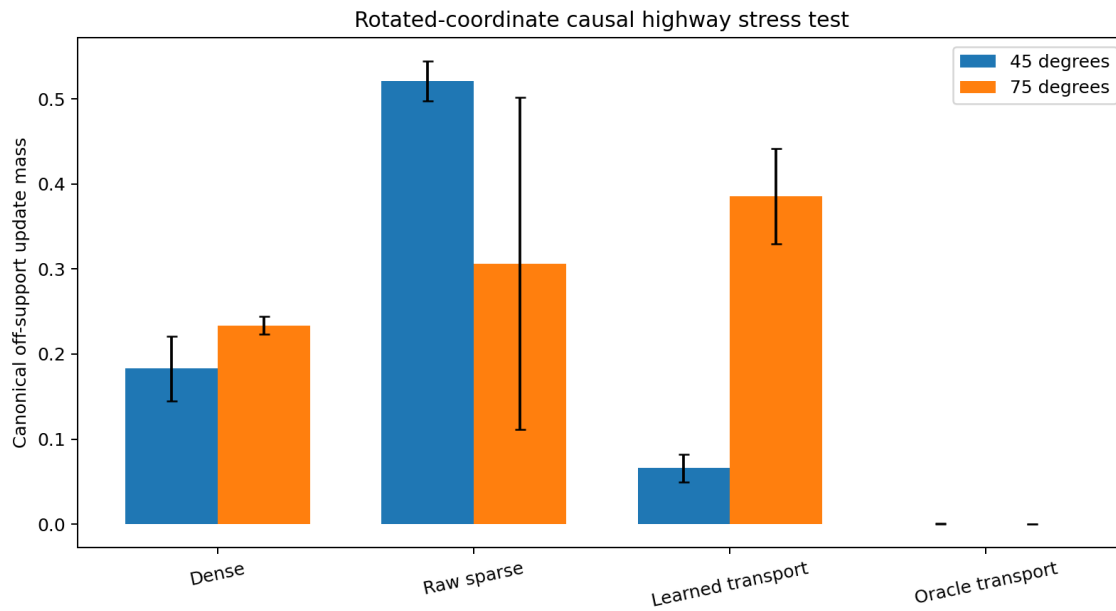


Figure 11: Canonical off-support update mass in the rotated-coordinate experiment. Oracle transport provides the architectural upper bound; raw sparse routing can be functionally effective yet causally wrong.

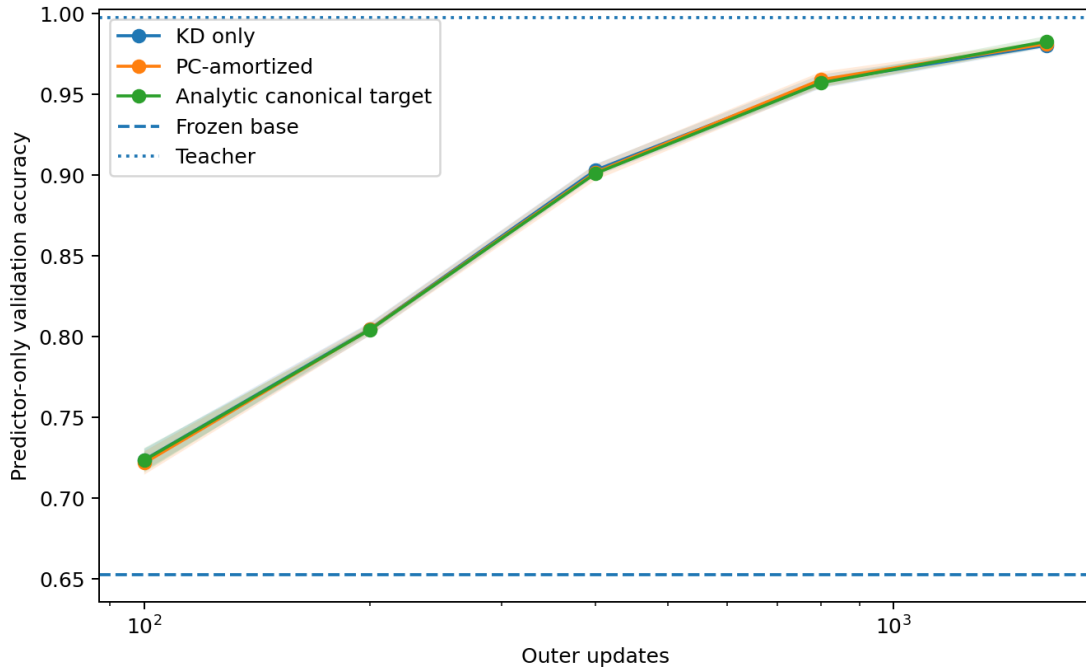


Figure 12: Learning curves from the canonical error-fibre proof-of-mechanism. The favorable coordinate system sharply reduced the deployment/clamped gap.

Condition	Predictor	Gap closure	Canonical MSE	Clamped
KD only, fixed fibres	0.9805	95.0%	4.720	0.9976
PC-amortized fixed fibres	0.9814	95.3%	3.382	0.9979
Direct analytic error target	0.9829	95.7%	2.472	0.9977

The frozen base accuracy was 0.6527 and teacher accuracy was 0.9977. The PC-amortized sidecar recovered about 95.3% of the available accuracy gap. Its accuracy advantage over output KD was small, but canonical error-coordinate MSE improved by about 28%.

In sequential remapping, only the responsible branch was writable. Unrelated output KL was exactly zero, while a parameter-similar shared predictor incurred large unrelated drift.

This experiment does not show that fibres emerge automatically. It shows that once a stable causal coordinate system and selective writes are available, the residual-PC and continual-learning mechanics can work cleanly.

11.5 S4: integrated toolkit validation

An installable package build was installed in isolation and exercised through 32 automated tests, CLI checks, packaged examples, and a custom integration suite.

11.5.1 Frozen-backbone distillation

A frozen two-layer MLP with a four-fibre terminal sidecar closed 99.15% of the teacher KL gap and 98.91% of the centered-logit gap across three seeds. Settlement energy was monotone on every outer update, and the predictor-versus-settled KL gap was below 10^{-6} .

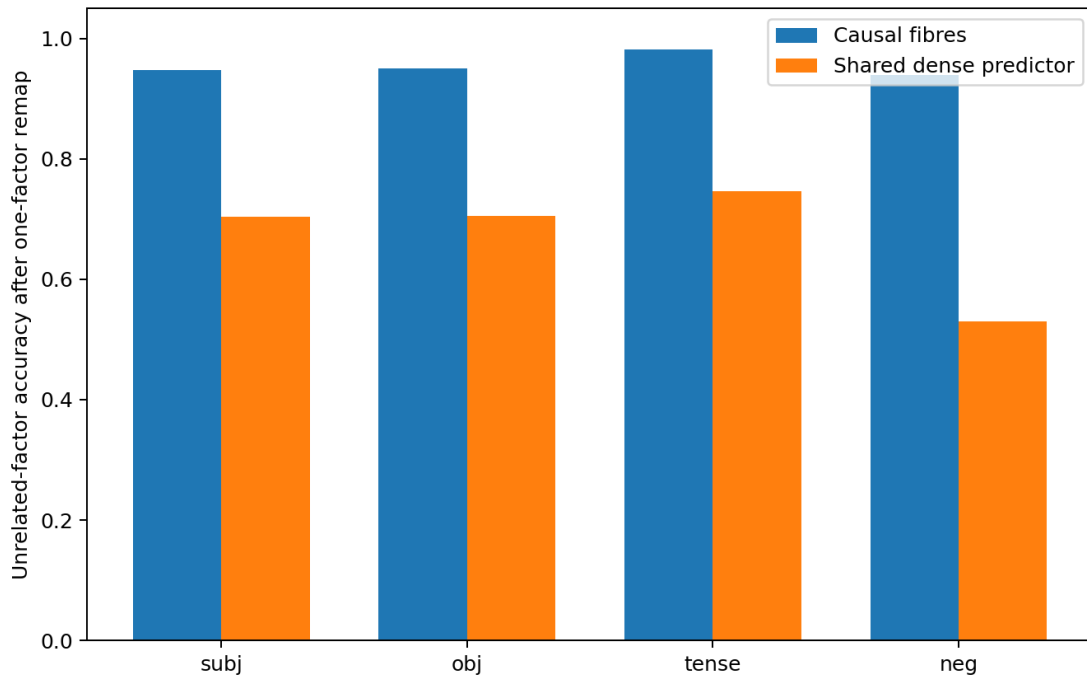


Figure 13: Continual adaptation in the canonical error-fibre experiment. Structurally disjoint branches preserve unrelated factors exactly under supplied supports.

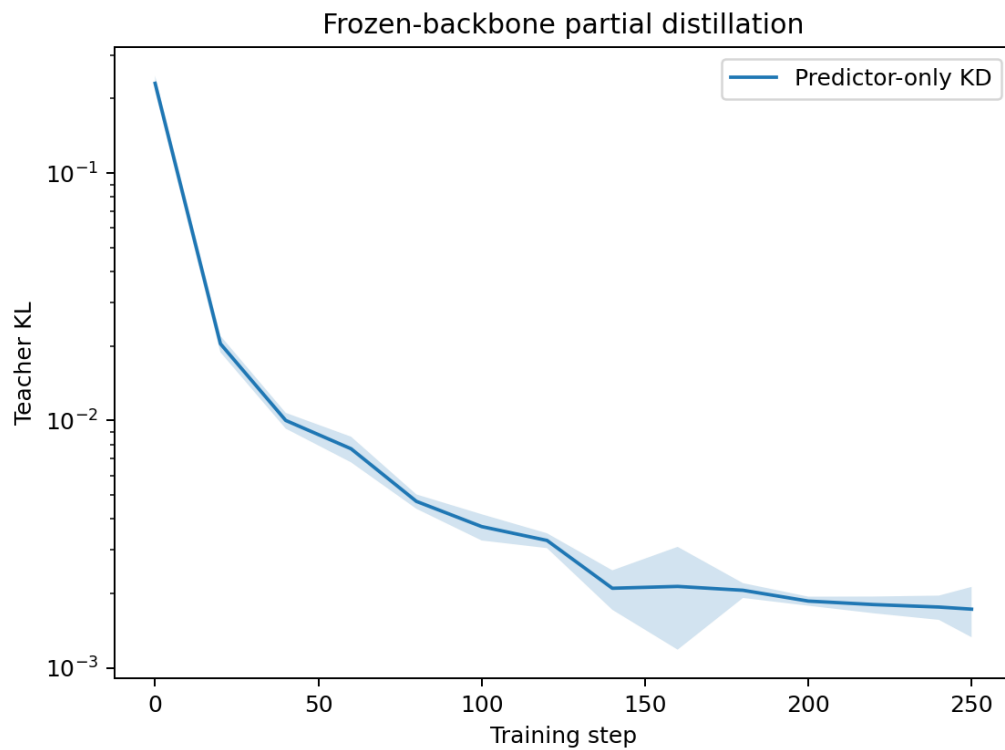


Figure 14: Held-out partial-distillation learning curve from the installed package integration suite.

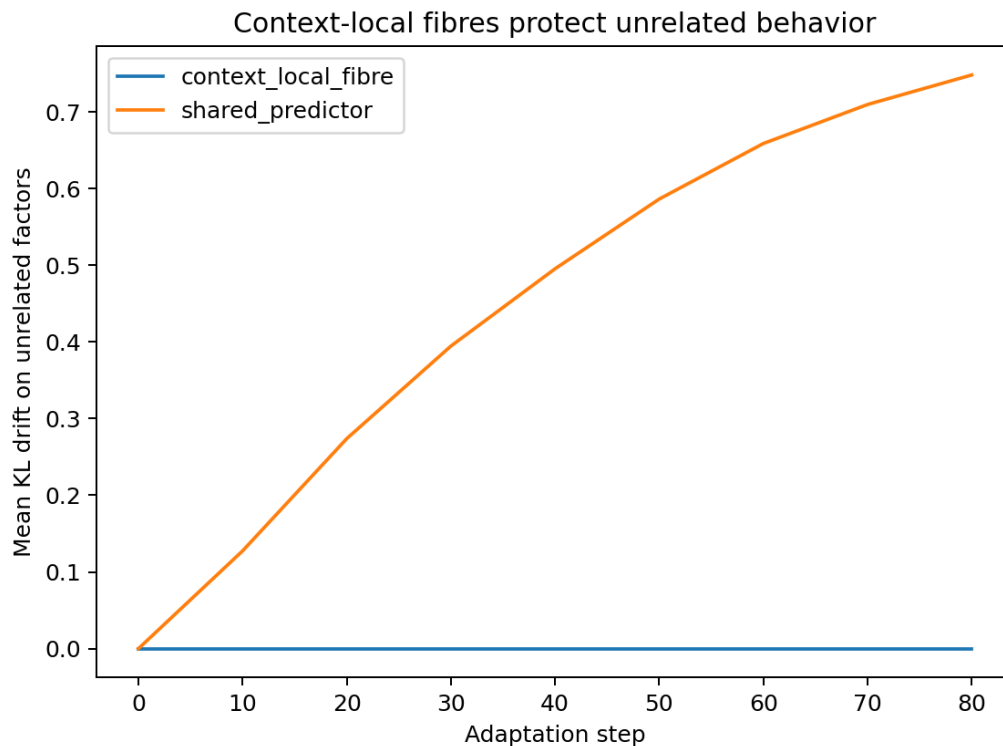


Figure 15: Installed-package continual-learning example. The supplied-support fibre model preserves unrelated outputs exactly; the shared predictor does not.

11.5.2 Context-local plasticity

Four independent fibre branches were compared with one shared predictor. After remapping only factor 0, the context-local model retained unrelated accuracy around 96.6% with zero unrelated KL drift. The shared predictor retained about 73.4% unrelated accuracy and had unrelated KL drift about 0.75.

11.5.3 Intervention fingerprints

Ablating each canonical fibre and evaluating all four factor-specific losses produced an almost perfectly diagonal intervention matrix.

11.5.4 Symbolic alignment

Four fibres, each with two symbolic states, were aligned to an eight-template store. Held-out top-1 retrieval rose from a mean of 16.4% before training to 100% after training, and the mutable store correctly accepted a ninth template.

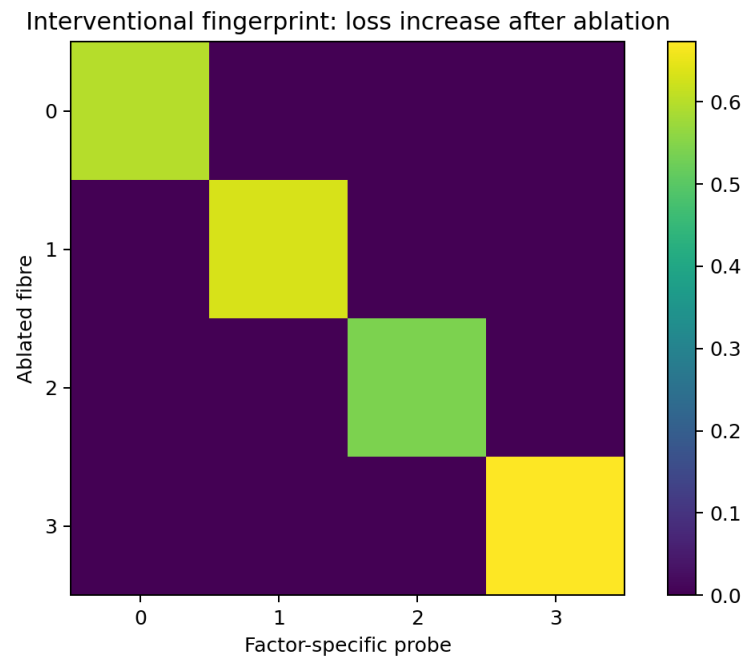


Figure 16: Fibre-by-probe intervention effects in the toolkit validation. Diagonal structure verifies the intervention and fingerprint APIs in the controlled case.

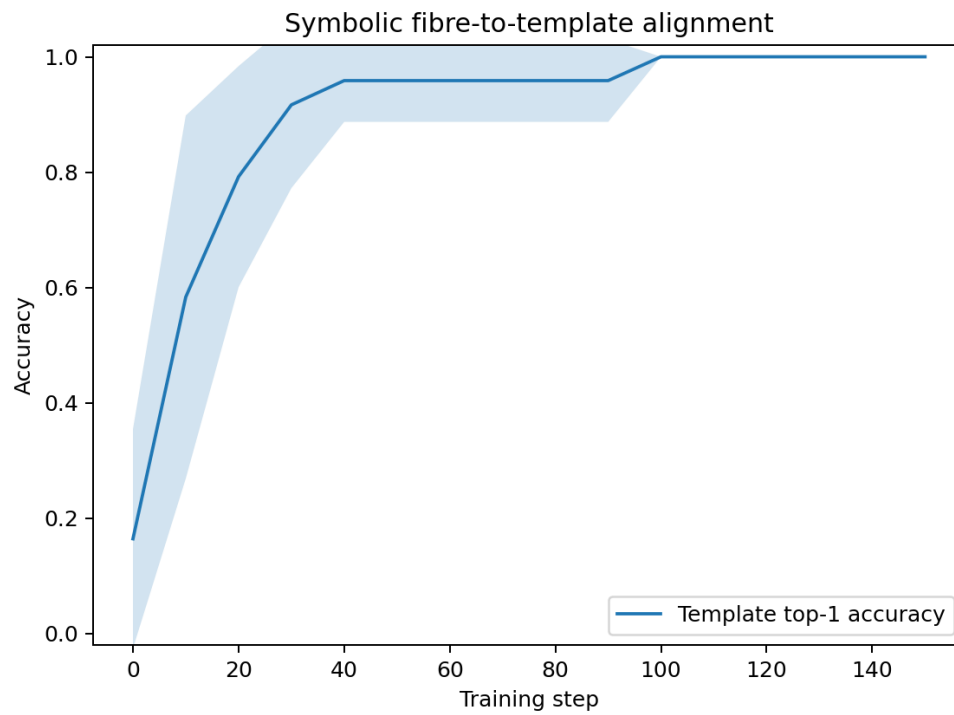


Figure 17: Held-out symbolic retrieval accuracy in the toolkit validation. This verifies the generic template-store and symbolic-fibre APIs, not a large symbolic reasoning benchmark.

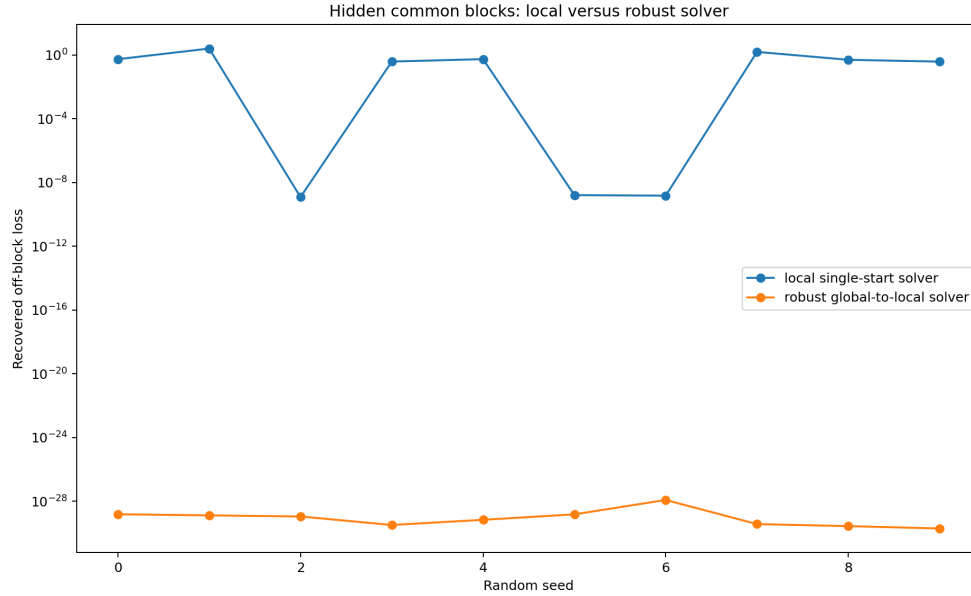


Figure 18: Exact hidden-block recovery with the local single-start and robust global-to-local solvers.

11.6 S5: robust joint block diagonalization

The operator stress test that defeated the local single-start optimizer was rerun with the robust global-to-local solver. Ten random families contained two hidden two-dimensional common invariant blocks with non-diagonal context-dependent internal matrices.

Solver	Mean block loss	Mean max angle	Recoveries below 0.1 rad
Local single-start	0.658393	1.09955	3/10
Robust global-to-local	1.97×10^{-29}	1.60×10^{-8}	10/10

At 10% symmetric operator noise, the robust solver still recovered all ten subspace pairs within 0.1 radians, with mean maximum angle about 0.047 radians.

A repeated-copy control used operators $Q \text{blockdiag}(A_c, A_c) Q^\top$. The fitted block loss was essentially zero, but the solver returned `repeated_or_underidentified`, a near-zero signature gap, and an oversized commutant. This is the desired scientific behavior: numerical fit without false uniqueness.

11.7 S6: evidence-gated package examples

The package includes sandbox examples that test the evidence-gated scientific sequence rather than assuming fibres from the start.

11.7.1 Frozen-read observability

Three read interfaces with identical 120-parameter residual probes achieved teacher-gap closures of approximately

$$0.184, \quad 0.576, \quad 1.000. \quad (142)$$

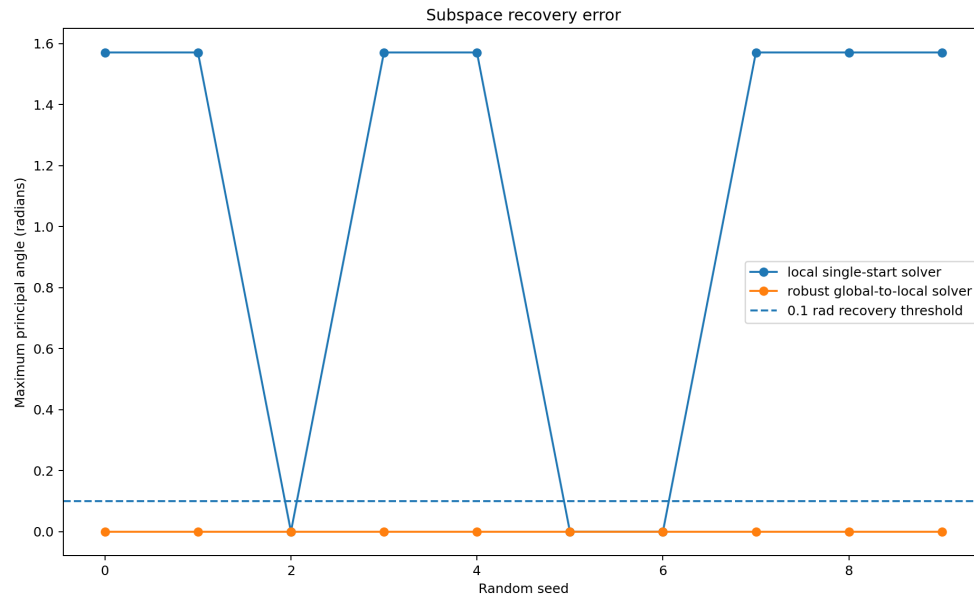


Figure 19: Principal-angle recovery of the generating fibre subspaces. The robust solver recovers all ten families to numerical precision.

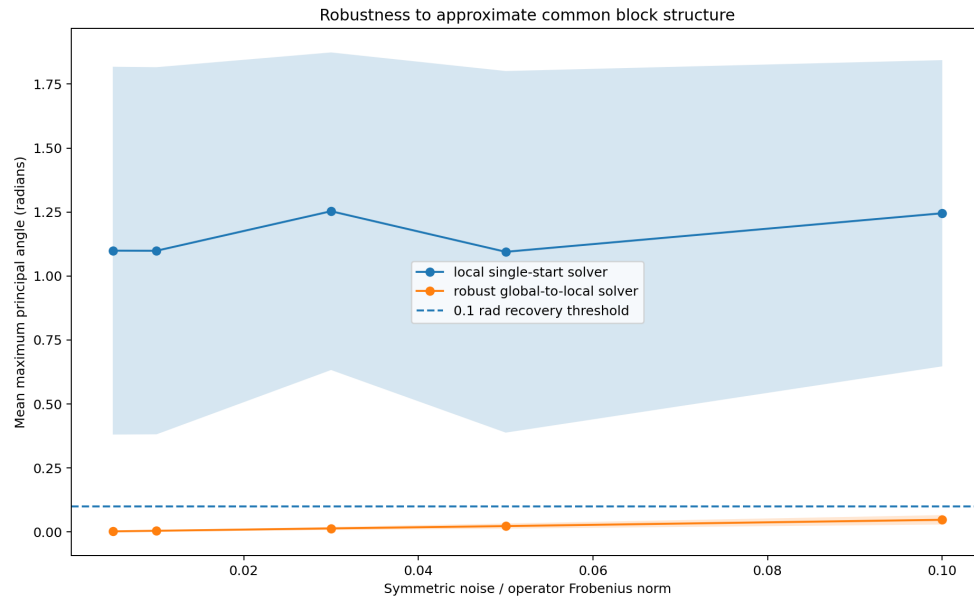


Figure 20: Robust joint block recovery as symmetric operator noise increases.

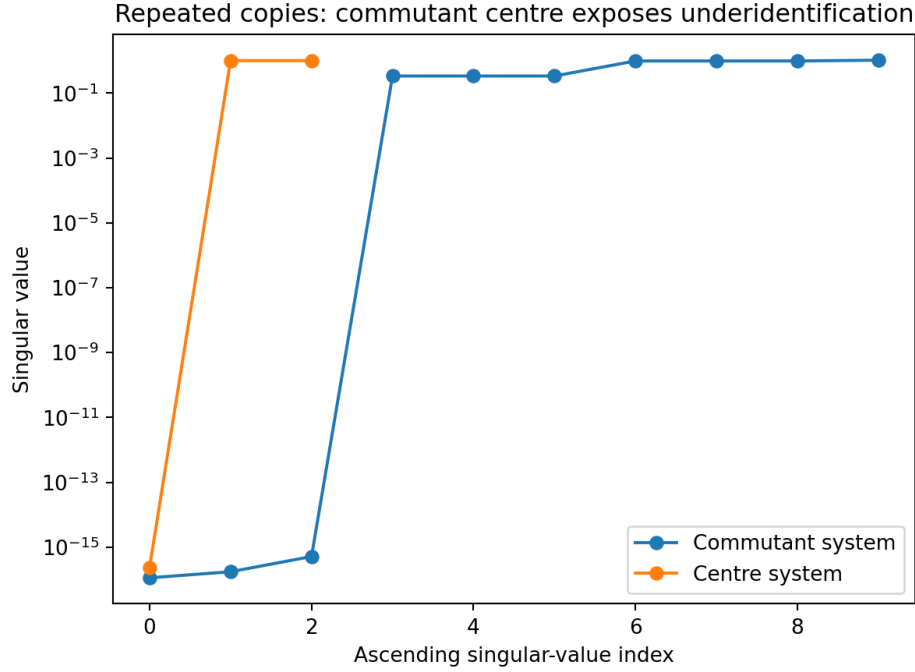


Figure 21: Commutant-centre spectrum in the repeated-copy control. The operator family admits perfect block fits but does not uniquely identify the two copies.

The late interface exposed essentially the full correction while the early interface did not. This illustrates why observability can dominate all later architectural choices.

11.7.2 Representation contest under superposition

A toy contest compared dense, SAE, grouped SAE, orthogonal, and contextual-bundle representations. The contextual bundle won the predictive teacher-gap criterion in this small run, with closure about 0.178, while the orthogonal representation achieved the best reconstruction among several compact contenders. The point is not the winner of one toy dataset. The point is that the software makes the representation hypothesis explicit and budget reportable.

11.7.3 Controlled substrate co-adaptation

A rank-2 low-rank update reduced held-out MSE from approximately 1.746 to 0.671 while reporting anchor drift of 0.213. The example verifies that freezing is now a recorded control and that carefully bounded substrate plasticity is a first-class experiment.

11.7.4 Deployment-honest settlement

The settlement example produced losses

$$L_{\text{predictor}} = 0.725, \quad L_{\text{free}} = 0.643, \quad L_{\text{constrained}} = 0.0313, \quad L_{\text{clamped}} = 0. \quad (143)$$

Free settlement using no new information helped modestly. Constrained settlement using a retrieved symbolic constraint helped dramatically. Teacher clamping is explicitly labeled an oracle diagnostic.

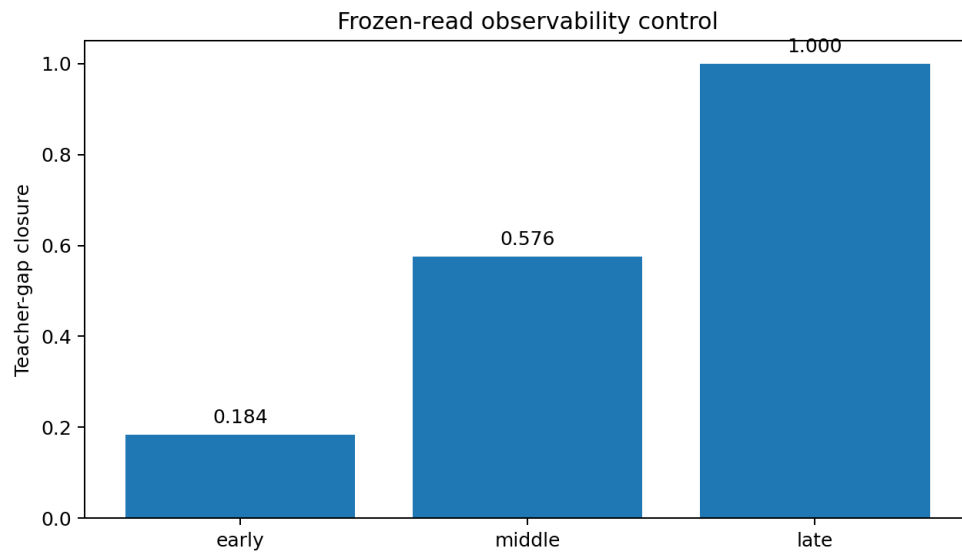


Figure 22: Frozen-read observability control from the installed package. The same residual capacity produces radically different teacher-gap closure at different interfaces.

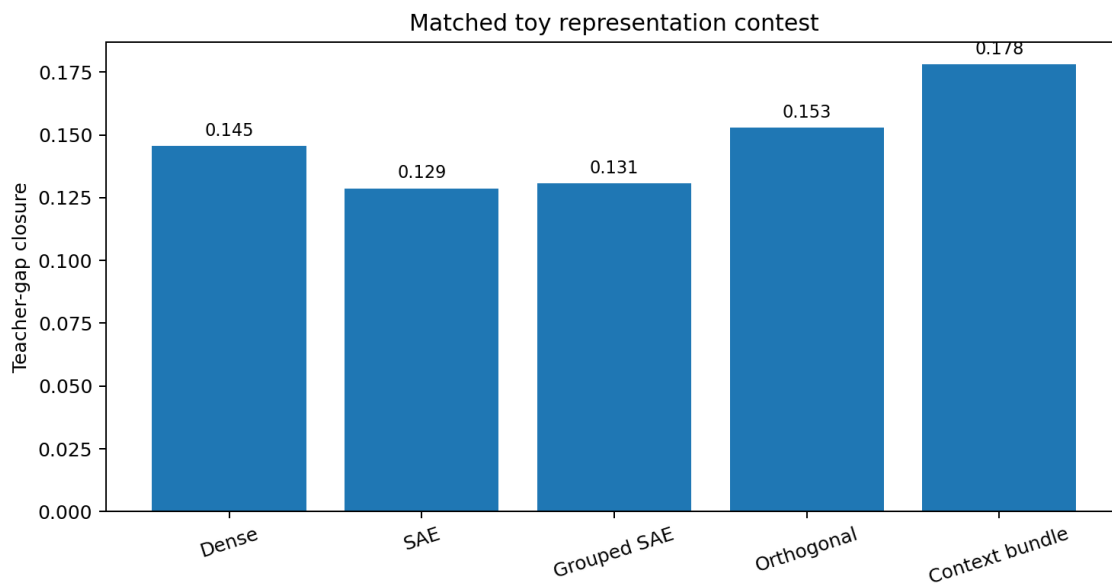


Figure 23: Toy representation contest shipped with the package. Orthogonal fibres are evaluated alongside dense and overcomplete alternatives rather than assumed.

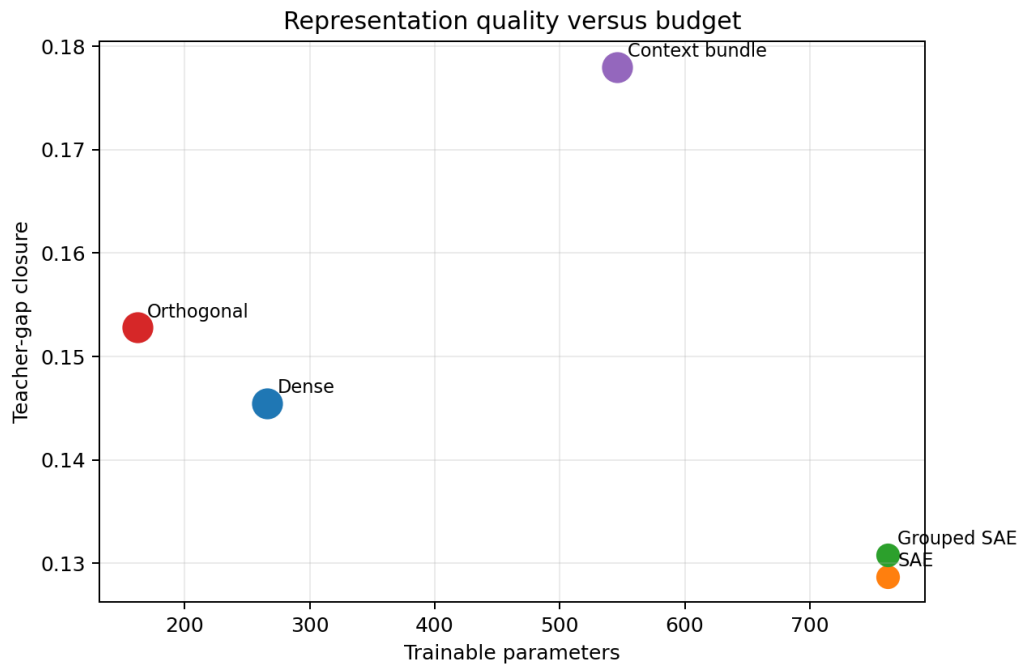


Figure 24: Teacher-gap closure versus trainable parameter budget in the toy representation contest. Marker size reflects mean active features.

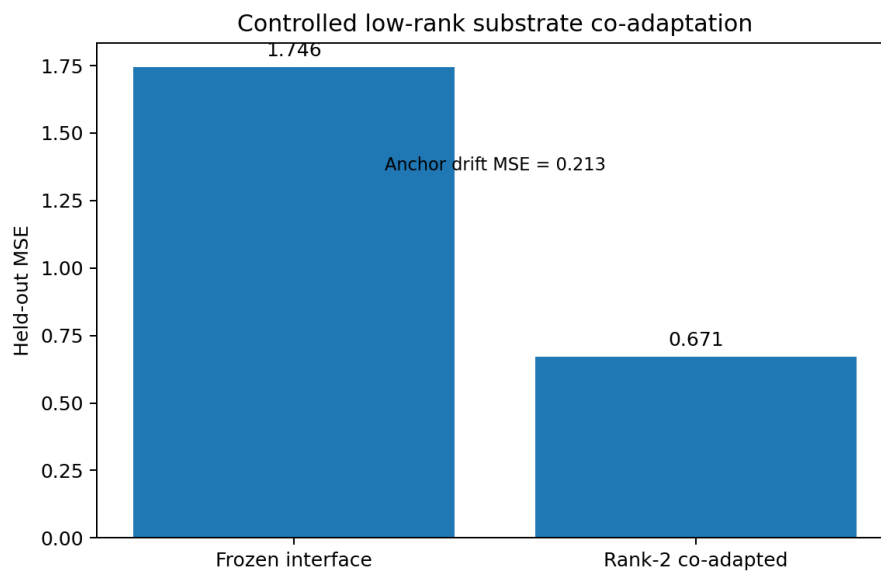


Figure 25: Controlled low-rank substrate co-adaptation in the package example. Improvement and anchor drift are reported separately.

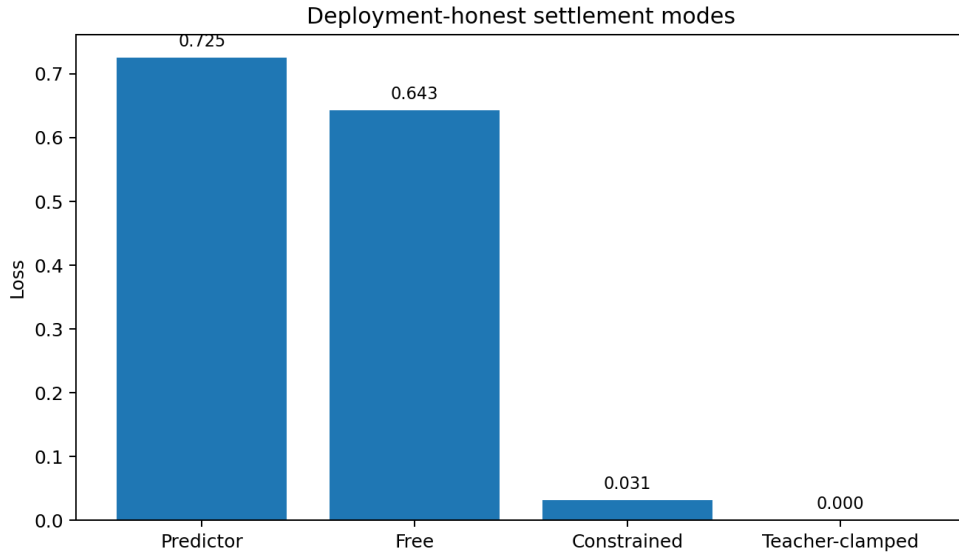


Figure 26: Predictor-only, free, constrained, and teacher-clamped modes are measured separately. Only constrained settlement receives new test-time information in this example.

11.7.5 Finite curriculum order

Three complete update sequences produced a maximum function-space order distance of approximately 1.66×10^{-3} . The example is intentionally small, but it validates direct finite-path evaluation rather than treating a pairwise commutator as a long-horizon guarantee.

11.8 What the sandbox evidence does and does not establish

The experiments support several narrow conclusions:

- a frozen sidecar can express and amortize a large teacher correction;
- short damped state-PC/ePC settlement can improve an explicitly defined error code;
- strict context-local plasticity can eliminate unrelated function-space drift when supports are supplied;
- task loss can improve under causally wrong credit, so intervention and support audits are necessary;
- transport must be identified separately from routing;
- static symbolic fibre alignment and retrieval are technically straightforward in a toy store;
- common invariant blocks can be discovered robustly at small dense residual-state scales when the operator family actually identifies them.

They do not establish that natural-language causal fibres emerge automatically, that ePC settlement outperforms backpropagation at matched wall-clock on realistic models, or that a symbolic cap will improve general reasoning. Those are the purposes of the staged GPU program.

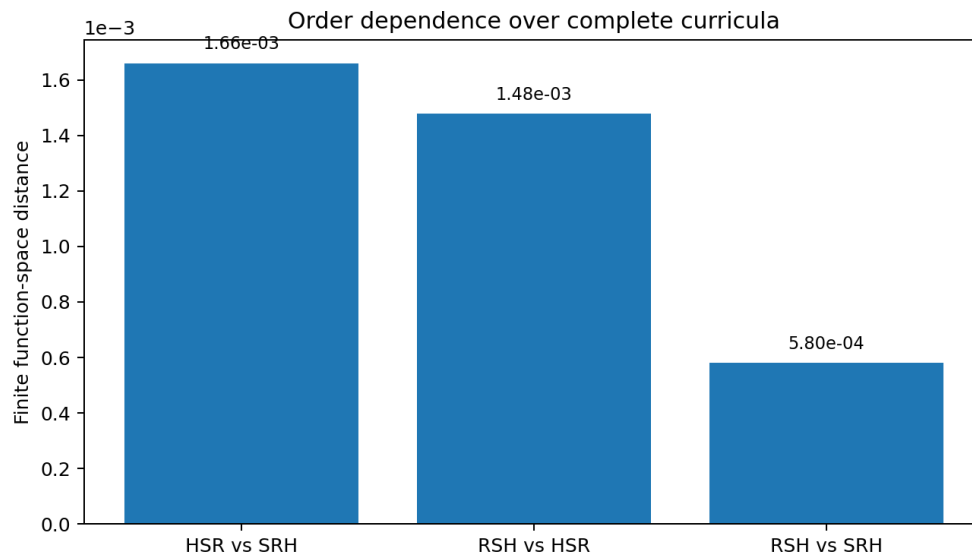


Figure 27: Finite function-space distances among complete curricula in the package example.

12 The causal-fibres software library

12.1 Purpose and design stance

The package is a PyTorch-first framework for *comparing and building structured adaptation systems*. It is not a package that assumes every model should be converted into an orthogonal PC Transformer. Its design separates:

$$\begin{aligned}
 &\text{architecture integration} \neq \text{representation model,} \\
 &\text{transient state inference} \neq \text{persistent parameter learning,} \\
 &\text{causal responsibility} \neq \text{coordinate transport,} \\
 &\text{task performance} \neq \text{causal certification,} \\
 &\text{oracle diagnostics} \neq \text{deployment evaluation.}
 \end{aligned} \tag{144}$$

12.2 Logical package organization

The source tree includes the following major namespaces:

Namespace	Main responsibilities
architectures	Generic hooks, functional adapters, and Hugging Face hidden-state capture.
representations	Dense bottlenecks, SAEs, grouped dictionaries, orthogonal fibres, contextual bundles, contests, and steering.
observability	Frozen-read probes, layerwise audits, and teacher-gap capacity curves.
substrate	Low-rank base plasticity, anchor trust regions, and alternating residual/base training.

fibres	Canonical state predictors, fibre banks, sidecars, plasticity controllers, pooling, and gauge tools.
solvers	State-gradient PC, generic preconditioned PC, LM/ePC settlement, and conjugate-gradient utilities.
credit	Exact oracle credit, factorization, routers, novelty residualization, transports, optional highways, and oracle blending.
operators	Hessian/Fisher/Gauss–Newton operators, sketches, JBD, commutant analysis, emergence metrics, and dictionary charts.
interventions	Ablations, replacements, perturbations, repair effects, and fingerprint tracking.
evaluation	Predictor/free/constrained/clamped settlement, matched compute, finite curricula, and novel-context routing.
certification	Graded module claims and underidentification handling.
symbolic	Template-store protocols, static and mutable stores, symbolic queries, attention, and feedback.
training	Distillation trainer, generic engine, update rules, losses, callbacks, schedules, and continual-learning utilities.
recipes	Staged structural recipes and stage gates.

12.3 Core protocols and data objects

The package uses small contracts so that host architectures and learning algorithms can vary independently. A site has a name, width, semantic kind, and write mode. A forward record contains model outputs and captured site states. A fibre state stores canonical values, priors, gates, masks, and metadata. A solve result stores the settled state, energy history, convergence and damping diagnostics. A step context passes these objects among trainers, credit sources, audits, and callbacks.

The architecture convention is that the final tensor dimension is the feature dimension while leading dimensions remain arbitrary:

Transformer	[batch, token, feature]
CNN	[batch, y , x , feature]
RNN	[batch, time, feature]
GNN	[node, feature]

This permits the same canonical-fibre and settlement code to operate tokenwise, spatially, temporally, globally, or per graph node.

12.4 Representation APIs

The representation layer contains five reference classes:

- `DenseResidualRepresentation;`
- `SparseAutoencoder;`
- `GroupedSparseDictionary;`
- `OrthogonalFibreRepresentation;`

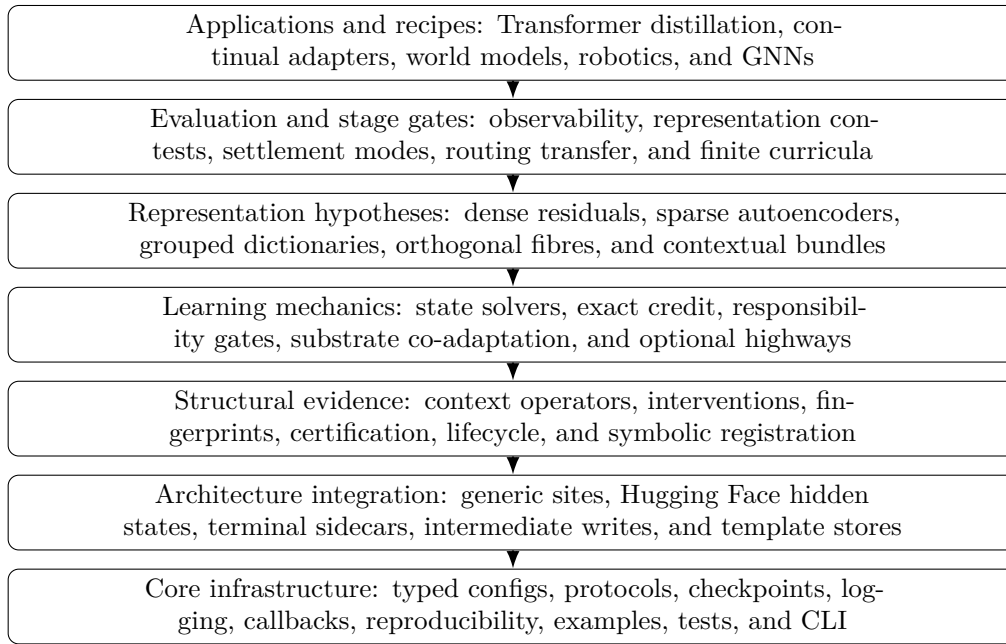


Figure 28: Logical organization of `causal-fibres`. Representation hypotheses, learning rules, and causal claims are separate layers rather than one monolithic Transformer-specific implementation.

- `ContextualFibreBundle`.

`RepresentationSteeringModel` pairs a representation with a prediction or steering head. `RepresentationSteeringContest` fits several candidates against the same base and teacher logits and reports reconstruction, active features, parameter counts, validation KL, and teacher-gap closure.

```

from causal_fibres.representations import (
    DenseResidualRepresentation,
    SparseAutoencoder,
    OrthogonalFibreRepresentation,
    RepresentationSteeringModel,
    RepresentationSteeringContest,
)

models = {
    "dense": RepresentationSteeringModel(
        DenseResidualRepresentation(hidden_size, 64),
        code_width=64,
        output_dim=vocab_size,
    ),
    "sae": RepresentationSteeringModel(
        SparseAutoencoder(hidden_size, 256, top_k=16),
        code_width=256,
        output_dim=vocab_size,
    ),
    "orthogonal": RepresentationSteeringModel(
        OrthogonalFibreRepresentation(hidden_size, 8, 8),
        code_width=64,
        output_dim=vocab_size,
    ),
}

```

```

    ),
}
report = RepresentationSteeringContest(models).fit(
    train_hidden,
    train_base_logits,
    train_teacher_logits,
    valid_hidden,
    valid_base_logits,
    valid_teacher_logits,
)

```

Listing 3: A minimal representation contest.

12.5 Observability and substrate APIs

FrozenReadProbe is the mandatory first control. It fits a dense or low-rank residual using exactly the proposed hidden read and output write. **TeacherGapCapacityCurve** repeats this across capacities; layerwise audits compare sites.

```

from causal_fibres.observability import FrozenReadProbe

probe = FrozenReadProbe(
    input_dim=hidden_size,
    output_dim=vocab_size,
    kind="low_rank",
    capacity=32,
    steps=500,
)
result = probe.fit(
    train_hidden,
    train_base_logits,
    train_teacher_logits,
    valid_hidden,
    valid_base_logits,
    valid_teacher_logits,
)
print(result.teacher_gap_closure)

```

Listing 4: Frozen-interface observability.

LowRankBasePlasticity wraps selected modules with dependency-free low-rank updates. **AnchorTrustRegion** measures behavior drift; **AlternatingResidualBaseTrainer** alternates residual and substrate phases without silently becoming unrestricted fine-tuning.

12.6 Fibre and sidecar mechanics

MultiSiteFibrePredictor combines several architectural sites into canonical states. **IndependentFibrePredictor** provides strict branch-local predictors for supplied-support controls. **LinearFibreBank** and constrained write dictionaries map fibres into output or residual-stream corrections. **TerminalFibreSidecar** provides a reversible terminal correction; setting its residual scale to zero exactly recovers the base.

Activation routing and persistent-plasticity routing are separate. A fibre may participate in the forward computation while being protected from a context update, or receive a small activation but a strong write-permission signal because it is causally novel.

12.7 Solvers and settlement modes

The solver protocol accepts an initial state and energy function. Implementations include no settlement, gradient PC, generic preconditioned PC, adaptive damped Levenberg–Marquardt/ePC settlement, and matrix-free conjugate gradient. Trust-ratio and backtracking diagnostics make inner-state failures visible.

The evaluation API names four modes:

- **PREDICTOR**: deployable amortized state;
- **FREE**: settlement without additional information;
- **CONSTRAINED**: settlement with retrieval, recurrence, rules, or other new constraints;
- **CLAMPED**: teacher-informed oracle diagnostic.

12.8 Exact credit and plasticity gating

ResponsibilityGradientGate applies responsibility weights after exact backpropagation and before the optimizer step. This is the conservative Stage 1 mechanism.

```
from causal_fibres.fibres import ResponsibilityGradientGate

gate = ResponsibilityGradientGate([
    parameters_for_fibre_0,
    parameters_for_fibre_1,
    parameters_for_fibre_2,
])

loss.backward()
metrics = gate.apply(responsibility)
optimizer.step()
```

Listing 5: Exact gradients with context responsibility.

The **credit** namespace also contains factor-error encoders, known-support and influence routers, soft top- k routing, novelty residualization, Procrustes transport initialization, path and metric transport losses, dense feedback controls, optional causal highways, and oracle blending.

12.9 Operator and intervention infrastructure

The package supplies dense and matrix-free Hessian, empirical-Fisher, and Gauss–Newton operators; randomized range estimates; Hutchinson traces; robust joint block diagonalization; dictionary-coordinate operator audits; context signatures; emergence ratios; and commutator matrices. Interventions operate on pure composition and loss functions, enabling state ablation or perturbation without hard-coding one architecture.

The robust `JointBlockDiagonalizer` uses approximate commutant and spectral initializers, multiple starts, signature-aware selection, and orthogonal-group refinement. It also reports repeated or underidentified operator families rather than arbitrarily certifying one of many equivalent decompositions.

12.10 Finite curricula and held-out routing

`FiniteCurriculumLoopEvaluator` clones a model or state, applies complete context sequences, and measures function-space distances. `NovelContextRoutingEvaluator` separates training-context and held-out-context support precision, recall, exact support, and off-support responsibility mass.

12.11 Certification and lifecycle

Module claims are evidence graded:

$$\text{candidate} \rightarrow \text{functional} \rightarrow \text{intervention-aligned} \rightarrow \text{causally identified} \rightarrow \text{consolidated}. \quad (145)$$

An explicit underidentified status is available. Lifecycle policies may propose spawn, split, merge, promote, freeze, or retire events. The package remains conservative: status and event proposals are implemented, while automatic tensor resizing and optimizer-state migration remain experimental engineering work.

12.12 Symbolic interface

A minimal `TemplateStore` protocol returns scores, keys, and values for a top- k query. Static and mutable stores support experiments; a future Hyperon/MORK plugin can implement the same protocol. Small-template retrieval is treated as an API test. Scientific evidence requires hard negatives, compositional retrieval, unseen arguments, and rule-mediated behavioral effects.

12.13 Configuration, stage gates, and reproducibility

The package defines this staged progression:

$$\begin{aligned} &\text{OBSERVABILITY} \rightarrow \text{REPRESENTATION_CONTEST} \rightarrow \text{GATED_PLASTICITY} \\ &\quad \rightarrow \text{SUBSTRATE_COADAPTATION} \rightarrow \text{PC_CONSTRAINTS} \\ &\quad \rightarrow \text{LEARNED_HIGHWAY} \rightarrow \text{SYMBOLIC_CAP} \end{aligned}$$

The recipe layer can block an unjustified stage, for example refusing a learned highway while exact residual credit remains cheap. Checkpoints include model, optimizer, scheduler, configuration, fibre registry, metrics, and random-number-generator state.

12.14 Installation, validation, and maturity

```
python -m pip install causal-fibres
causal-fibres doctor
causal-fibres smoke-test
causal-fibres list-representations
```

The release passed 54 automated tests, source and example compilation, all supplied configurations, clean wheel and source-distribution installation, and runnable examples. It remains an alpha research package. Production limitations include distributed large-model training, native intermediate writes

for every model family, automatic structural tensor mutation, and a networked Hyperon/MORK backend.

13 Application to partial Transformer distillation

13.1 Meaning of partial distillation

The goal is not to replace a competent Transformer immediately. Let

$$a^0 = F_0(x), \quad a^T = F_T(x) \quad (146)$$

be base and teacher logits. A structured residual learns

$$a^S = a^0 + R_\phi(h^0, c), \quad (147)$$

where h^0 consists of selected base states and c is optional context information. The residual may later contain transient PC states, selective plasticity, symbolic constraints, or context-dependent transports. The base remains intact or receives only controlled low-rank updates until the residual has passed staged tests.

Partial distillation has four objectives:

1. preserve the broad competence and healthy internal geometry of the base;
2. learn a smaller correction space whose structure can be inspected and intervened upon;
3. determine which representation family best supports selective continual adaptation;
4. create a stable interface through which retrieval, symbolic frames, rules, or persistent state may constrain test-time inference.

13.2 Stage 0A: frozen observability

For each candidate layer or pooled read, fit an ordinary BP residual using the proposed write path. Produce teacher-gap closure curves versus rank, parameter count, and wall clock. Include tokenwise and pooled variants. A failure at this stage is a failure of the interface, not evidence against predictive coding.

Required baselines include:

- dense residual MLP;
- low-rank residual or LoRA-style adapter;
- linear probe plus steering when meaningful;
- the same terminal write planned for later sidecars.

A reasonable initial gate is that at least one matched-size ordinary residual should close more than half the available teacher KL gap and expose the factors or behaviors later structural stages intend to manipulate.

13.3 Stage 0B: matched representation contest

Using exact gradients and the same read/write interface, compare:

1. dense low-rank residual;
2. SAE plus steering;
3. grouped overcomplete sparse dictionary;
4. orthogonal fibre representation;
5. context-conditioned fibre bundle;
6. optional variants with low-rank substrate co-adaptation.

Match or explicitly report:

- total parameters ever trained;
- deployment parameters;
- training and deployment FLOPs or wall clock;
- code width, active features, and group count;
- teacher-gap closure and validation KL;
- reconstruction or predictive error;
- intervention selectivity, redundancy, and feature stability;
- finite curriculum-order sensitivity.

The representation contest does not prove causality. It selects useful structural candidates for later intervention and continual-learning tests.

13.4 Stage 1: exact-gradient gated plasticity

On the best two or three representation classes, retain exact residual gradients and compare:

1. ordinary exact-gradient adaptation;
2. exact gradient with supplied support masks;
3. exact gradient with learned responsibility gates;
4. replay, regularization, and parameter-isolation baselines.

Training contexts and held-out contexts must be separated. The learned gate should be evaluated for support precision, support recall, off-support responsibility mass, current-task improvement, and unrelated drift. The main question is whether support estimation itself transfers, not whether a router can memorize task IDs.

13.5 Stage 2: controlled substrate co-adaptation

Compare the frozen winner with selected-module low-rank co-adaptation. Use anchor data and a behavioral trust region. Measure whether modest substrate plasticity improves observability, representation score, intervention selectivity, signature separation, and continual retention without excessive base drift.

This phase directly addresses the possibility that the frozen backbone was never pressured toward the desired structure. It should occur before interpreting failure of PC settlement as a fundamental limitation.

13.6 Stage 3: state-PC and ePC residual constraints

Only after a representation and support mechanism have passed should transient residual states be introduced. Compare:

- feed-forward predictor only;
- state-gradient PC;
- damped Gauss–Newton ePC settlement;
- matched-depth recurrent adapter;
- free settlement with no new information;
- constrained settlement with retrieval, symbolic rules, recurrence, or persistent memory.

A typical residual energy is

$$E(z) = L_{\text{task}}(a^0 + Wz) + \frac{\lambda_{\text{pred}}}{2} \|z - \mu_{\psi}(h^0)\|_{\Pi}^2 + \frac{\lambda_{\text{abs}}}{2} \|z\|^2 + E_{\text{constraints}}(z). \quad (148)$$

The amortized outer objective should include direct predictor-only task loss:

$$L_{\text{outer}} = L_{\text{deploy}}(\mu_{\psi}) + \lambda_{\text{amort}} \|\mu_{\psi} - \text{sg}(z^*)\|^2 + \lambda_{\text{anchor}} L_{\text{anchor}}. \quad (149)$$

Report all settlement modes separately and normalize improvements by additional training and deployment compute.

13.7 Stage 4: optional learned highway

Proceed only if exact gradients are a genuine bottleneck or if learned credit transfer is itself the hypothesis. Compare on unseen contexts:

1. exact gradient;
2. exact gradient plus learned plasticity gate;
3. dense learned feedback;
4. learned causal highway;

5. oracle-blended transition.

The highway earns its place only if it improves locality, held-out-context support or credit transfer, asynchronous operation, biological plausibility, or computational cost. Matching the oracle on training contexts is not enough.

13.8 Stage 5: symbolic or memory constraints

Begin with a static, nontrivial template library and hard negatives. Evaluate neural-to-symbolic retrieval, symbolic rule inference, and selective symbolic-to-neural feedback. A useful test hides a surface decision, retrieves or infers the relevant frame, feeds it into a designated residual mechanism, and measures correct behavioral recovery with minimal unrelated change.

The scientific comparison must include ordinary retrieval augmentation or memory attention. A symbolic cap is valuable only if explicit structured inference improves behavior, continual retention, or transfer.

13.9 Stage 6: intermediate writes and partial absorption

Once the terminal sidecar passes the earlier gates, introduce one intermediate write at a time:

$$\tilde{h}_\ell = h_\ell^0 + W_\ell z, \quad \tilde{h}_L = F_{L:\ell+1}^0(\tilde{h}_\ell). \quad (150)$$

Compare against the terminal sidecar at matched residual capacity. Successful and stable residual mechanisms may later be absorbed into low-rank base updates or replacement blocks using a homotopy that preserves task behavior, intervention fingerprints, symbolic bindings, and finite curriculum properties.

13.10 First six-layer GPU matrix

A compact first program should use the same six-layer synthetic-grammar infrastructure but change the scientific order.

Stage	Conditions	Decisive question
0A	Layerwise dense and low-rank BP residuals	Is the correction observable at the proposed sites?
0B	Dense, SAE, grouped SAE, orthogonal, contextual bundle	Which structure wins at matched budget?
1	Exact gradient, supplied mask, learned gate, replay	Does selective plasticity reduce unrelated drift and transfer to held-out contexts?
2	Frozen versus rank-constrained co-adaptation	Does limited substrate plasticity improve structural feasibility?
3	Predictor, state-PC, damped ePC, recurrent control	Does settlement add value at matched compute?
4	Optional exact-gated versus learned highway	Does learned credit generalize or solve a real bottleneck?
5	Static symbolic/memory constraints	Does structured test-time information improve behavior?

6	One intermediate write, then selective absorption	Can the successful residual be integrated without losing its audits?
---	---	--

13.11 Primary metrics and kill conditions

Primary metrics are teacher-gap closure, predictor-only validation, training and deployment cost, observability by site, active feature budget, intervention selectivity, held-out support transfer, unrelated drift, finite curriculum-order distance, representation stability, and anchor drift.

Important kill conditions include:

- no ordinary residual can recover the correction from the proposed interface;
- a structured model loses to dense and SAE baselines without a compensating continual-learning or symbolic benefit;
- learned supports fail on held-out contexts;
- PC settlement helps only under teacher clamping;
- free settlement does not beat matched recurrent computation;
- a highway matches the oracle only on seen contexts and offers no cost or locality advantage;
- symbolic retrieval succeeds but rule feedback has no selective behavioral effect.

14 Neural-symbolic cap compatibility

14.1 What “cap amenable” means

A continuous residual is cap amenable when its internal state supports stable, typed, bidirectional interactions with an external reasoning or memory system. Useful properties include:

- a small number of stable fibre IDs rather than a diffuse basis over the entire backbone;
- explicit interventional meanings for fibre perturbations;
- a typed template or frame distribution associated with each fibre;
- controlled symbolic-to-neural feedback;
- shared cores for durable concepts and shells for local context;
- explicit mediator fibres for relations and rules;
- persistence of fibre-template binding across continual adaptation.

14.2 Symbolic head as registration, not post hoc interpretation

The symbolic projection should be trained jointly with the residual. Given template keys p_j and values v_j , use contrastive alignment

$$\mathcal{L}_{\text{sym}} = - \sum_{j \in P^+(z_q)} \log \frac{\exp((W_q^{\text{sym}} z_q)^\top p_j / \tau)}{\sum_k \exp((W_q^{\text{sym}} z_q)^\top p_k / \tau)} + \lambda_v \mathcal{L}_{\text{value}}. \quad (151)$$

This avoids assuming that a clean neural cluster automatically corresponds to a well-typed symbolic atom. The mapping is a learned and testable retrieval operation.

14.3 Symbolic interventions

A symbolic assertion can act as a constraint on a fibre:

$$E_{\text{rule},q} = \lambda_{\text{rule}} \left\| W_q^{\text{sym}} z_q - v_{\text{rule}} \right\|^2. \quad (152)$$

Free settlement then translates the rule into a neural correction. Selectivity should be measured as

$$\text{Sel}(j) = \frac{\text{effect on intended outputs}}{\epsilon + \text{effect on unintended outputs}}. \quad (153)$$

A symbolic cap is useful only if interventions remain selective and stable across contexts.

14.4 Asynchronous consolidation

At larger scale, high residual surprise combined with low template similarity can trigger an asynchronous consolidation path:

1. capture context, fibre state, and error fingerprint;
2. parse or mine a candidate symbolic template off the critical path;
3. add the template to a persistent store;
4. replay or revisit examples so the symbolic projection aligns with the new template;
5. bind the stable template to the relevant fibre or mediator.

The fast path should continue without waiting for consolidation.

15 Broader architectural applications

The same framework applies whenever a family of contexts induces different update operators on a modular state space.

15.1 Recurrent world models

Fibres can be recurrent state subspaces corresponding to object dynamics, latent causes, goals, or environmental regimes. Controlled actions provide natural interventions.

15.2 Robotics and skill learning

Fibres may represent actuators, perceptual causes, reusable skills, or contact modes. Context-local plasticity can protect old skills while mediator fibres represent genuine coordination.

15.3 Graph neural networks

Fibres may correspond to node types, relation types, or mechanism-specific subgraphs. Site transports become message-passing coordinate maps.

15.4 Multimodal systems

One canonical fibre can have visual, linguistic, auditory, and motor charts. Transport consistency becomes a direct cross-modal alignment criterion.

15.5 Differentiable programs and non-gradient learners

The context operator need not be a Hessian. It can be the Jacobian of an update map, an empirical update covariance, or a discrete endomorphism difference. The common abstraction remains a sparse, approximately multiplicity-free invariant decomposition of context-conditioned updates.

16 Failure modes and open questions

16.1 Failure modes

Observed failure	Likely cause	Response
High clamped performance, weak deployment	predictor cannot infer settled state; target leakage in evaluation	add direct deployment loss; audit reads; reduce latent nonuniqueness
Raw latent collapse	scale/write gauge; overly strong contraction pressure	normalize writes; add absolute state prior; report function-space contributions
Energy descends but causal support is wrong	dense or misaligned credit	measure off-support mass; train against oracle; add intervention transport anchors
Router correct, update wrong	transport identified poorly	Procrustes initialization; path and metric losses; true interventions
One fibre dominates	weak competition, identical modules, early advantage	soft load balancing; reserve fibres; detuning; lifecycle split
Many duplicate fibres	no capacity cost or duplicate audit	fingerprint/read/write redundancy penalty; merge policy
No stable decomposition	insufficient context variation or dense interactions	active interventions; mediator fibres; abandon hard factorization if red regime
Small block loss but unstable semantics	repeated isomorphic modules or commuting degeneracy	inspect signature gap and commutant; add anchors or detune

Good current task, poor retention	cross-support curvature or modewise schedule mismatch	distinguish metric and connection channels; add modularity or replay as appropriate
Symbolic head ignored	task can be solved without it; weak integration	increase alignment/feedback gradually; require symbolic intervention tests
PC too slow	too many settlement steps or expensive metrics	amortize predictor; use $K \leq 4$; block/low-rank metrics; adaptive stopping

16.2 Open mathematical questions

- Can nonlinear causal-fibre emergence be given a global convergence theorem under useful assumptions?
- How should block sizes be discovered rather than supplied?
- Can matrix-free approximate-commutant solvers scale to hundreds or thousands of residual dimensions?
- What is the best finite-sample confidence interval for signature gaps and emergence ratios?
- How should transport holonomy be separated into benign local chart changes and genuine mediator structure?
- When do symbolic anchors improve causal identifiability rather than impose a misleading ontology?
- Can fibre lifecycle operations preserve optimizer state and causal fingerprints under distributed training?
- Which operator family is most predictive of continual-learning benefit: Fisher, Gauss–Newton, exact Hessian, update covariance, or a hybrid?

16.3 Open empirical questions

- Does a canonical error-fibre sidecar close a substantial teacher gap on the six-layer grammar model at matched wall-clock?
- Does damped state-PC/ePC settlement improve sample efficiency or internal code quality beyond ordinary residual KD?
- Can learned causal routing approach supplied-support retention without excessive intervention cost?
- Do operator-derived fibres agree with counterfactual grammar factors and symbolic templates?
- Does a static symbolic cap improve compositional generalization or only interpretability?
- Do the same mechanisms survive natural-language scale, longer contexts, and less explicit causal labels?

- Can mature fibres be transferred across base-model updates more cheaply than retraining the residual?

17 Conclusion

The approach embodied in `causal-fibres` is no longer a claim that pretrained Transformers naturally contain a small set of orthogonal causal fibres. It is a staged methodology for finding out which structural adaptation model, if any, is useful and identifiable.

The first responsibility of the methodology is falsification. A frozen-read observability control asks whether the proposed interface exposes the correction. A matched representation contest asks whether orthogonal fibres, overcomplete sparse features, contextual transports, dense residuals, or controlled substrate adaptation provide the best account. Exact gradients with responsibility-conditioned persistent plasticity test the continual-learning hypothesis without confounding it with learned credit. Finite curriculum-order tests supplement local commutators. Deployment-honest settlement separates predictor, free, constrained, and teacher-clamped behavior. A highway and symbolic cap appear only after they have a distinct empirical role to play.

The mathematical core remains valuable. Common invariant subspaces characterize one clean regime; signature gaps, perturbation bounds, multiplicity, and commutant structure explain identifiability. Overcomplete dictionary coordinates and contextual fibre bundles broaden that model toward superposition and context-dependent charts. Observability and controllability define interface feasibility. Intervention fingerprints, support transfer, off-support gradients, mixed Hessian blocks, and finite curriculum maps connect representation structure to selective plasticity and forgetting.

The initial Transformer objective is therefore modest but consequential:

Preserve a competent base; learn and compare small structured residuals; permit limited substrate co-adaptation when needed; add short predictive-coding settlement only when it helps; and expose a stable, auditable interface through which continual learning, retrieval, memory, or symbolic constraints can act.

A positive result need not be “orthogonal causal fibres emerge.” It may be that grouped SAE features plus exact-gradient gates are the best continual substrate, or that a contextual bundle is required, or that a dense residual remains superior. The package is scientifically useful when it reports those outcomes honestly. Its strongest promise is not a predetermined architecture but a disciplined route from broad Transformer competence toward smaller, inspectable, selectively plastic, and eventually neural-symbolically constrained adaptation systems.

A Detailed algorithms

A.1 Factor-supervised residual training

Inputs:

```
frozen base F0
optional teacher FT
selected read sites S
fibre predictors mu_q
canonical-to-output writers B_q
optional transport maps T_lq
```

```

    number of settlement steps K

for minibatch (x, y, context c):
    # Frozen computation
    with no_grad():
        h_sites, base_logits = F0.forward_with_sites(x, S)
        teacher_logits = FT(x) if FT is not None else None

    # Amortized state
    prior = predictor(h_sites, c)
    z = prior

    # Short clamped settlement
    for k in range(K):
        gates = soft_router(h_sites, c, z)
        residual_logits = writer(z, gates)
        logits = base_logits + residual_logits

        external = CE_or_KD(logits, y, teacher_logits)
        predictor_energy = 0.5 * norm_Pi(z - prior)**2
        absolute_energy = 0.5 * lambda_abs * norm(z)**2
        registration = multi_site_registration(h_sites, z, T_lq)
        symbolic = symbolic_constraint(z) if enabled else 0
        energy = external + predictor_energy + absolute_energy \
            + registration + symbolic

        metric = build_state_metric(z, logits, energy)
        task_credit = exact_or_highway_credit(z, logits, teacher_logits)
        local_credit = grad(predictor_energy + absolute_energy \
            + registration + symbolic, z)
        z = trust_region_step(z, metric, task_credit + local_credit)

    settled = stop_gradient(z)

    # Persistent outer learning
    deploy_logits = base_logits + writer(prior, soft_router(...))
    deploy_loss = CE_or_KD(deploy_logits, y, teacher_logits)
    amort_loss = norm_Pi(prior - settled)**2
    settled_loss = CE_or_KD(base_logits + writer(settled, ...),
        y, teacher_logits)

    outer_loss = deploy_loss + lambda_amort * amort_loss \
        + lambda_settled * settled_loss \
        + gauge_and_anchor_losses()

    outer_loss.backward()
    apply_context_local_plasticity_mask()
    optimizer.step()

periodically:
    audit_interventions()
    estimate_context_operators()
    update_joint_block_structure()

```

```
calibrate_transports()
log_predictor_and_clamped_metrics_separately()
```

Listing 6: Factor-supervised terminal-sidecar training.

A.2 Learned causal highway with oracle annealing

```
alpha = 1.0

for batch in training_stream:
    oracle = exact_task_gradient_in_canonical_fibre_coordinates(batch)

    factor_errors = error_encoder(batch.output_error,
                                  batch.context,
                                  batch.hidden_sites)
    novel_errors = novelty_filter(factor_errors, shared_memory)
    responsibility = router(batch.hidden_sites,
                             batch.context,
                             factor_errors,
                             influence_uncertainty)
    highway = transport_to_fibres(responsibility, novel_errors)

    credit_loss = whitened_mse(highway, stop_gradient(oracle)) \
                  + cosine_penalty(highway, oracle)
    update_router_transport_and_error_encoder(credit_loss)

    used_credit = alpha * oracle + (1 - alpha) * highway
    apply_residual_update(used_credit)

    if held_out_audit_passes(
        cosine_min=0.8,
        off_support_max=0.1,
        energy_fraction_min=0.9,
        transport_condition_max=condition_limit,
        no_deploy_regression=True,
    ):
        alpha = max(alpha_min, alpha - anneal_step)
    else:
        alpha = min(1.0, alpha + recovery_step)
```

Listing 7: Causal credit distillation.

A.3 Online structure update

```
for each structural audit interval:
    unexplained = task_error - sum(predicted_fibre_effects)

    if persistent_metric_norm(unexplained) > spawn_threshold:
        propose_spawn(initial_direction=leading_unexplained_mode)

    for fibre q:
        if fingerprint_is_multimodal(q) and \
```



```

        internal_JBD_loss_after_split(q) < split_threshold:
            propose_split(q)

for fibre pair (q, r):
    if fingerprints_close(q, r) and \
        read_subspaces_close(q, r) and \
        write_subspaces_close(q, r) and \
        held_out_merge_cost(q, r) < merge_threshold:
            propose_merge(q, r)

for fibre q:
    if stable_across_contexts(q) and low_curvature(q):
        propose_promote_to_shared_core(q)
    if unique_ablation_value(q) < retire_threshold:
        propose_retire(q)

require human-readable diagnostics and checkpoint
before materializing any tensor-resizing event

```

Listing 8: Conservative fibre lifecycle policy.

B Metrics and audits

B.1 External and deployment metrics

Metric	Definition or purpose
Teacher gap closure	fraction of accessible teacher improvement captured by the residual
Predictor-only loss	target-free deployment performance
Teacher-clamped loss	residual capacity and credit diagnostic
Amortization gap	predictor-only loss minus clamped loss
Anchor KL	behavior drift from frozen base on protected data
Matched wall-clock score	external performance under equal elapsed compute
Settlement iterations	number of latent steps needed per example or batch

B.2 State and gauge metrics

Metric	Definition or purpose
Raw effective rank	entropy rank of concatenated latent covariance
Fibre-balanced rank	rank after equalizing per-fibre total variance
Function-space contribution	RMS or covariance of $B_q z_q$
Scale ratio	maximum to minimum fibre contribution scale
Energy monotonicity	fraction of steps satisfying $E_{k+1} \leq E_k$
Trust ratio	actual divided by predicted energy reduction
Damping trajectory	adaptive LM damping through settlement

B.3 Causal and continual-learning metrics

Metric	Definition or purpose
Repair responsibility	loss reduction under a small fibre intervention
Off-support credit	fraction of task update outside intended or measured support
Intervention selectivity	intended effect divided by unintended effect
Fingerprint conditioning	minimum singular value or sparse restricted singular value
Context signature gap	minimum distance between operator-use profiles
Emergence ratio	signature separation divided by cross-block coupling and noise
Update commutator	function-space difference between AB and BA updates
Sequential-joint gap	difference between curriculum training and mixture training
Forgetting	old-context loss increase relative to its best previous value
Confusion graph	context pairs with both support overlap and noncommutativity

B.4 Transport metrics

Metric	Definition or purpose
Intervention transport error	mismatch between transported canonical perturbation and observed local effect
Path-consistency defect	difference between alternative transport compositions
Metric distortion	$\ T^\top G_{\text{local}} T - G_{\text{canonical}}\ _F$
Condition number	numerical stability and message amplification of T
Cycle holonomy	product of transports around a local-representation loop

B.5 Symbolic metrics

Metric	Definition or purpose
Template precision/recall	retrieval against annotated or mined templates
Cluster separation	intra-template versus inter-template distance in key space
Fibre-template mutual information	specialization and binding strength
Symbolic intervention selectivity	functional effect of clamping a symbolic value
Binding stability	persistence across seeds, contexts, and base updates
Round-trip consistency	neural-to-symbolic-to-neural reconstruction or effect

C Suggested configurations for the six-layer program

C.1 Stage 0A observability

```

observability:
  read_sites: [embedding, block_0, block_2, block_3, block_4, block_5]
  pooling: [tokenwise, masked_mean, learned_attention]
  residual_types: [linear, low_rank, dense_mlp]
  capacities: [4, 8, 16, 32, 64]
  write_path: final_logits
  report: [teacher_gap_closure, validation_kl, parameters, wall_clock]

```

C.2 Stage 0B representation contest

```

representations:
  - dense:
      code_width: 64
  - sae:
      code_width: 256
      top_k: 16
  - grouped_sae:
      code_width: 256
      groups: 8
      overlap: true
  - orthogonal:
      fibres: 8
      fibre_dim: 8
  - contextual_bundle:
      canonical_width: 64
      context_rank: 8
matching:
  enforce_parameter_report: true
  enforce_active_feature_report: true
  match_training_steps: true
  add_matched_wall_clock: true

```

C.3 Stage 1 exact-gradient plasticity

```

credit:
  mode: oracle_gated
plasticity:
  controls:
    - ungated_exact_gradient
    - supplied_support
    - learned_responsibility
    - replay
routing_evaluation:
  held_out_contexts: true
  metrics: [precision, recall, f1, exact_support, off_support_mass]
curricula:
  orders: [AB, BA, ABC, CBA, random_permutations]

```

C.4 Stage 2 substrate co-adaptation

```

substrate:
  frozen_control: true
  modules: [block_2.attn, block_3.mlp, block_5.mlp]
  ranks: [2, 4, 8]
  anchor_trust_region:
    metric: symmetric_kl
    radius: 0.02
  alternating_schedule:
    residual_steps: 4
    base_steps: 1

```

C.5 Stage 3 PC constraints

```

settlement:
  modes: [predictor, free, constrained, clamped]
  solvers: [none, gradient_pc, damped_lm]
  steps: [1, 2, 4]
  adaptive_damping: true
  absolute_state_penalty: 0.01
controls:
  matched_recurrent_adapter: true
  matched_deployment_compute: true

```

C.6 Optional Stage 4 highway

```

highway:
  enabled_only_if_exact_gradient_bottleneck: true
  controls: [oracle, oracle_gated, dense_feedback, causal_highway]
  train_contexts: [A, B, C]
  held_out_contexts: [D, A_plus_C]
  oracle_blend: 1.0
  anneal_on:
    credit_cosine: 0.8
    off_support_mass: 0.1
    held_out_support_f1: 0.7

```

D Software API reference snapshot

Class or function	Role
<code>FrozenReadProbe</code>	ordinary residual observability falsifier
<code>TeacherGapCapacityCurve</code>	closure versus residual capacity
<code>DenseResidualRepresentation</code>	dense bottleneck candidate
<code>SparseAutoencoder</code>	overcomplete sparse representation candidate
<code>GroupedSparseDictionary</code>	possibly overlapping feature groups
<code>OrthogonalFibreRepresentation</code>	direct-sum fibre candidate
<code>ContextualFibreBundle</code>	canonical code with context-dependent local charts
<code>RepresentationSteeringContest</code>	matched teacher-residual representation comparison

SAEFeatureSteering	mandatory SAE steering baseline
LowRankBasePlasticity	selected-module low-rank substrate update
AnchorTrustRegion	bound substrate behavior drift
AlternatingResidualBaseTrainer	alternate residual and base plasticity
MultiSiteFibrePredictor	combine several sites into canonical states
IndependentFibrePredictor	strict branch-local predictor for supplied-support controls
LinearFibreBank	grouped fibre writes
TerminalFibreSidecar	reversible terminal correction
ResponsibilityGradientGate	gate exact persistent gradients by responsibility
GradientPCSolver	state-gradient settlement
LevenbergMarquardtPCSolver	damped metric-conditioned settlement
AutogradOracleCredit	exact residual-gradient oracle
CausalHighwayCredit	optional routed, transported, novelty-filtered credit
OracleBlend	gradual credit distillation
SoftTopKRouter	soft sparse routing
InfluenceRouter	influence- and uncertainty-based responsibility
SiteTransportBank	local coordinate transports
AutogradGaussNewtonOperator	context-conditioned operator
JointBlockDiagonalizer	robust common invariant block discovery
audit_dictionary_operators	overcomplete operator-coordinate audit
FibreInterventionEngine	ablation and perturbation audits
FingerprintTracker	interventional effect profiles
FiniteCurriculumLoopEvaluator	complete update-order comparison
NovelContextRoutingEvaluator	held-out support generalization
ModuleCertifier	evidence-graded structural claims
LifecycleManager	conservative structure and status proposals
SymbolicFibreHead	per-fibre template queries and feedback
PartialDistillationTrainer	concrete residual distillation path
stage_gate	enforce the evidence-gated hypothesis ladder

E Reproducibility and artifact inventory

The analyses summarized in this report were generated from the following families of artifacts:

- the six-layer R8/R9 activity-energy distillation report;
- theoretical notes on causal continual learning, the coupling dilemma, causal evidence accounting, causal error highways, and transfer rigidity;
- the staged predictive-coding residual research plan;
- CPU sandbox reports for the initial sidecar, rotated-coordinate transport, and canonical error fibres;
- the package integration validation;
- the robust joint-block-diagonalization showcase;

- the package examples, configurations, and validation logs.

The source bundle accompanying this document includes the ASCII LaTeX file and the figure assets used to compile the PDF. The software package itself is distributed separately as a wheel, source archive, and release bundle.

F Selected bibliography

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